



Geometrically nonlinear isogeometric analysis of smart piezoelectric FG plates considering thermal effects of piezoelectric stress and dielectric constants

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ABSTRACT

In this paper, the isogeometric analysis (IGA) integrated with a simple first-order shear deformation theory (S-FSDT) with only four variables per node is extended to study geometrically nonlinear natural frequency, nonlinear static bending and transient dynamic response of two types of smart piezoelectric functionally graded plates (PFGPs) in thermal environment. The neutral surface is introduced to avoid the stretching-bending coupling effect. The nonlinear strain–displacement relation of the plates is obtained by combining the von Karman strains with the S-FSDT. To investigate the nonlinear mechanical behavior of PFGPs in thermal environment, closer to the real situation, an augmented piezoelectric constitutive equation considering the thermal effects of piezoelectric stress and dielectric constants is used. The accuracy and effectiveness of developed approach are demonstrated through numerical experiment with comparison. We additionally investigate the variation of temperature distribution on geometrically nonlinear response of PFGPs suffering thermo-electro-mechanical load.

1. Introduction

Composite smart piezoelectric structures are widely used in a variety of engineering applications, see e.g., [1,2] and references therein, because of their self-diagnostic, self-adapted and fast response performances. These structures refer to sticking (or embedding) the piezoelectric materials on the surface (inside) of beams, plates, shells and other structures, so that when the external environment changes, they can respond rapidly according to their physical characteristics. In real application, the smart piezoelectric structures are often subjected to large mechanical, electrical, and thermal loads, presenting nonlinear mechanical behavior. Studying the nonlinear behavior of smart piezoelectric structures to improve the accuracy in structural analysis and design is thus necessary. Generally, the nonlinear analysis of smart piezoelectric structures can be classified into three categories, i.e., the geometric nonlinearity [3], the reversible [4,5] and irreversible [6] piezoelectric material nonlinearities under large electric fields, and the nonlinearity with consideration of both geometric and piezoelectric material nonlinearities [7,8].

Studies on geometrically nonlinear response of composite smart piezoelectric structures have been conducted, and several approaches including analytical, semi-analytical, experimental and numerical methods have been introduced in the literature. For instance, by using a high-order shear deformation theory (HSDT), Yang et al. [9,10] derived semi-analytical solutions that account for the nonlinear vibration and bending behaviors of piezoelectric functionally graded plates (PFGPs) with initially thermo-electro-mechanical stress. Fu et al. [11] applied Galerkin technique to develop an analytical solution to investigate the nonlinear buckling, frequency and dynamic stability of piezoelectric functionally graded (FG) beams in thermal environment in terms of the incremental harmonic balance method. Using the aerofoil as the engineering background, Zhang et al. [12] analyzed nonlinear transverse vibration of a cantilevered piezoelectric laminated composite plate subjected to the transverse, in-plane and voltage excitations. Using the Ritz energy method, Ansari et al. [13] presented an analytical solution for the nonlinear post-buckling analysis of piezoelectric nanocomposite cylindrical shell structure. A theoretical solution for predicting the nonlinear dynamic behavior of the

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piezoelectric FG cylindrical shells subjected to electrical, thermal, mechanical and aerodynamic loads was presented by Rafiee *et al.* [14,15]. Assuming that the material property of piezoelectric material (PZT-5A) are linear functions of temperature. Shen [16] presented the nonlinear thermal bending response of PFGPs, while Huang and Shen [17], Xia and Shen [18] investigated the nonlinear vibration and dynamic response of PFGPs in thermal environment.

In terms of employing numerical methods, Shegokar and Lal [19,20] applied the finite element method (FEM) to analyze stochastic nonlinear bending and dynamic response of piezoelectric FG beams under thermo-electro-mechanical loads with random material property. Behjat [21] estimated the nonlinear stability and free vibration of PFGPs with the aid of the FEM. His research shows that with the increase of electric and mechanical loads, the deviation between the nonlinear solution and the linear solution becomes larger. Both the occurrence of large deformations and rotations are considered, Zhang *et al.* [22–25] used a fully geometrically nonlinear FEM to assess the nonlinear static and dynamic response of smart piezoelectric structures. A similar study for nonlinear frequency, dynamic response of PFGPs in thermal environment was performed by Fakhari *et al.* [26] but taking a HSDT-based FEM. To avoid reduction of the accuracy caused by element distortion, Nourmohammadi and Behjat [27] alternatively applied a mesh-free radial point interpolation method (RPIM) for nonlinear static bending response of PFGPs. Recently, Phung *et al.* [28] used the isogeometric analysis (IGA) to model the transient nonlinear response of PFGPs under thermo-electro-mechanical loads and the transient nonlinear control [29] of piezoelectric composite plates.

In addition to that, a few studies about the nonlinear control of piezoelectric structures have been reported in the literature. For instance, Fu *et al.* [30] analyzed the nonlinear control of damaged piezoelectric laminated plates. Panda and Ray [31] analyzed the active constrained layer damping of geometrically nonlinear transient response of PFGPs in thermal environment. Fakhari *et al.* [32] employed two control methods including a classical displacement-velocity feedback control method and robust H2 control method to achieve the control of PFGPs under thermo-mechanical loads. Zhang *et al.* [33] employed various nonlinear shell theories to study the active shape and vibration control of piezoelectric composite structures. Zhu *et al.* [34] presented a semi-analytical solution for the nonlinear vibration control of the porosity-dependent piezoelectric conical sandwich panels with elastic foundation.

In the above-mentioned nonlinear studies of piezoelectric structures in thermal environment, the material properties of functionally graded materials (FGMs) and piezoelectric layers were both regarded as temperature-independent, e.g., see Refs. [9,10]. In other references, except for Ref. [16], the temperature dependence of FGMs is considered, but the piezoelectric layer is taken to be temperature-independent. Most of existing studies treated the piezoelectric stress and dielectric coefficients of the piezoelectric materials at different temperatures as constants. However, it is evident from Ref. [35] that these coefficients are actually sensitive to temperature. Although the active control is not involved in this paper, it should be mentioned that in the vibration control of the smart piezoelectric plate structures, if two piezoelectric layers are symmetrically distributed on the upper and lower surfaces of the substrate, the control may become unstable [36–39] because of the stretching-bending coupling effect. This defect can be avoided by placing the two piezoelectric layers on the same side of the substrate. Meanwhile, when the structures are used in thermal environment, the two piezoelectric layers can be placed on the low temperature side to protect the performance of the piezoelectric materials from being affected by temperature as much as possible.

Due to the high temperature resistance of FGMs, we use the FGMs as the substrate to investigate the nonlinear free vibration, bending and dynamic response of PFGPs with two different struc-

tures in thermal environment. The IGA integrated with the simple first-order shear deformation theory (S-FSDT) [40,41] is adopted, and further extended to fulfill the given problem, the selected numerical model lies in the fact that it is because of the advantages of low-cost, less degrees of freedom (only four variables per node), naturally free of shear-locking [40]. Different from previous studies, the material properties of FGMs are considered to be temperature dependent, as well as the influences of temperature on piezoelectric stress and dielectric constants are introduced. To the best knowledge of the present authors, there are no such studies of geometrically nonlinear analysis for PFGPs by using the S-FSDT based IGA considering the thermal effect of piezoelectric coefficients. Additional goal of this work is to assess the nonlinear free vibration, static bending and dynamic response analysis of PFGPs in thermal environment closer to the real situation.

The rest of the paper is structured as follows. Mathematical model for PFGPs is derived in Section 2. Section 3 is the derivation process of isogeometric finite element motion equations for PFGPs. In Section 4, the solution procedures nonlinear natural frequencies, nonlinear static bending and nonlinear dynamic are developed. We provide four numerical examples in Section 5 to verify the accuracy and effectiveness of present S-FSDT based IGA method, and the temperature distribution and geometrically nonlinear response of PFGPs under thermo-electro-mechanical loads are then analyzed.

2. Mathematical model

Let us particularly consider the piezoelectric functionally graded plates with the size of $a \times b \times h_t$ (length $\times$ width $\times$ thicknesses) whose configurations are shown in Fig. 1. The total thickness of the plate $h_t = h_f + 2h_p$, where h_f is the thicknesses of the FGMs, and h_p is the thicknesses of each piezoelectric layer. Two piezoelectric layers of the S1 plate structure in Fig. 1(a) are symmetrically distributed on the upper and lower surfaces of the FGP. Whilst in the S2 plate structure in Fig. 1(b), the piezoelectric layers are both placed on the lower surfaces (low temperature side) of the FGP. V_u and V_l are the electric voltages applied on the upper and lower piezoelectric layers, respectively.

2.1. Thermal analysis

The effective material properties of the FGMs along the thickness direction can be described as

$$P(z, T) = (P_c - P_m)V_c(z) + P_m, \quad (1)$$

The volume fraction of ceramic materials is defined as

$$V_c(z) = \left(\frac{1}{2} + \frac{z}{h_f} \right)^n, \quad (2)$$

where the subscript symbols m and c represent the metal and ceramic, respectively; P_c and P_m are the corresponding material properties such as mass density (ρ), Young's modulus (E) and various other properties of ceramic and metal; n is the gradient index.

The temperature dependent material properties of FGMs are expressed as [42]

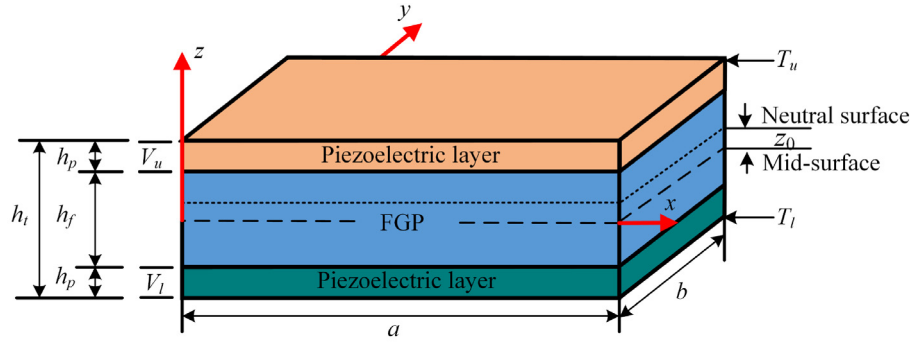
$$P_{c/m} = P_0(P_{-1}T^{-1} + 1 + P_1T + P_2T^2 + P_3T^3), \quad (3)$$

where P_{-1}, P_0, P_1, P_2 and P_3 are the coefficients of temperature T and are unique to each constituent.

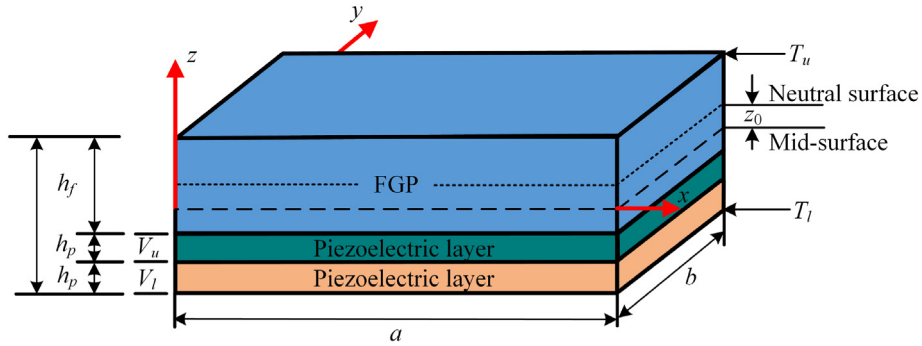
In thermal analysis, the temperature gradient along the thickness can be solved by a steady heat state transfer equation

$$-\frac{d}{dz} \left(k(z) \frac{dT(z)}{dz} \right) = 0 \quad (4)$$

where



(a) S1 plate



(b) S2 plate

Fig. 1. Schematic configuration of two types of piezoelectric functionally graded plates.

$$k(z) = \begin{cases} k_{pu} & h_f/2 < z < h_f/2 + h_p \\ k_f(z) & -h_f/2 < z < h_f/2 \\ k_{pl} & -h_f/2 - h_p < z < -h_f/2 \end{cases}, \quad (5)$$

$$T(z) = \begin{cases} T_{pu}(z) & h_f/2 < z < h_f/2 + h_p \\ T_f(z) & -h_f/2 < z < h_f/2 \\ T_{pl}(z) & -h_f/2 - h_p < z < -h_f/2 \end{cases}, \quad (6)$$

in which, k_{pu} and k_{pl} denote the thermal conductivity of the upper and lower piezoelectric layers, respectively; k_f is the thermal conductivity of FGMs which can be calculated from Eqs. (1)–(3); $T(z)$ is the temperature distribution function of the plates.

It is assumed that the temperature of the top and bottom surfaces of PFGP are T_u and T_l . Hence, Eq. (4) can be calculated by introducing the thermal boundary conditions

$$T_{pu}\left(\frac{h_f}{2} + h_p\right) = T_u, T_{pl}\left(-\frac{h_f}{2} - h_p\right) = T_l, \quad (7)$$

and the continuity conditions

$$T_{pu}(h_f/2) = T_f(h_f/2) = T_c, \quad (8a)$$

$$T_{pl}(-h_f/2) = T_f(-h_f/2) = T_m, \quad (8b)$$

$$k_{pu} \frac{dT_{pu}(z)}{dz} \Big|_{z=h_f/2} = k_f \frac{dT_f(z)}{dz} \Big|_{z=h_f/2}, \quad (9a)$$

$$k_{pl} \frac{dT_{pl}(z)}{dz} \Big|_{z=-h_f/2} = k_f \frac{dT_f(z)}{dz} \Big|_{z=-h_f/2}. \quad (9b)$$

Based on Eqs. (4)–(9), $T_{pu}(z)$, $T_{pl}(z)$, $T_f(z)$ can be obtained through a uniform form as [11]

$$T(z) = T_l + \frac{T_u - T_l}{\int_{-h_i/2}^{h_i/2} \frac{1}{k(z)} dz} \int_{-h_i/2}^z \frac{1}{k(z)} dz. \quad (10)$$

For the S2 structure, a similar solution process can be carried out for the temperature gradient. The volume fraction of ceramic materials is

$$V_c(z) = \left(\frac{1}{2} + \frac{z - d}{h_f}\right)^n, \quad d = \frac{h_t}{2} - \frac{h_f}{2}. \quad (11)$$

2.2. Kinematics description of plates using S-FSDT

Basically, the simple FSDT (S-FSDT) is derived from the conventional FSDT with two assumptions (see [40,41] and references therein) i.e., the transverse displacement $w_0(x, y)$ in the standard FSDT is divided into bending w_b and shear w_s , that is, $w_0(x, y) = w_b + w_s$; and the rotation variables are expressed using the partial derivatives of the transverse bending components only, that is, $\beta_x = -\partial w_b / \partial x$ and $\beta_y = -\partial w_b / \partial y$. Therefore, we can find out that the five variables ($u_0, v_0, w_0, \beta_x, \beta_y$) in the standard FSDT are reduced to four in the S-FSDT (u_0, v_0, w_0, w_b), where u_0 and v_0 represent the displacements of the mid-surface of the plate in the x and y directions, respectively; β_x and β_y are the rotations of the mid-surface around the y and x axes, respectively.

To avoid the stretching-bending coupling effect, the neutral surface [43] is introduced into S-FSDT [41]. The mechanical displacement formulation is denoted as

$$\begin{aligned} u(x, y, z) &= u_0(x, y) - (z - z_0) \partial w_b / \partial x \\ v(x, y, z) &= v_0(x, y) - (z - z_0) \partial w_b / \partial y, \\ w(x, y, z) &= w_b(x, y) + w_s(x, y) \end{aligned} \quad (12)$$

where z_0 is the distance between the mid-surface and neutral surface of the plates as sketched in Fig. 1, which can be expressed as

$$z_0 = \frac{\int_{-h_t/2}^{h_t/2} z E(z) dz}{\int_{-h_t/2}^{h_t/2} E(z) dz}. \quad (13)$$

By combining von Karman strains with the S-FSDT, the nonlinear strain-displacement relation of the plates can then be expressed as

$$\tilde{\epsilon} = \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ r_{xy} \\ r_{xz} \\ r_{yz} \end{Bmatrix} = \begin{Bmatrix} u_x + \frac{1}{2}(w_x)^2 \\ v_y + \frac{1}{2}(w_y)^2 \\ u_y + v_x + w_x w_y \\ w_x + u_z \\ w_y + v_z \end{Bmatrix}. \quad (14)$$

Substituting Eq. (12) into Eq. (14), yields

$$\tilde{\epsilon} = \begin{Bmatrix} \epsilon_m^0 + \epsilon_{nl} \\ \mathbf{0} \end{Bmatrix} + \begin{Bmatrix} -(z - z_0) \epsilon_b^0 \\ \epsilon_s^0 \end{Bmatrix}, \quad (15)$$

with

$$\epsilon_m^0 = \begin{Bmatrix} u_{0,x} \\ v_{0,y} \\ u_{0,y} + v_{0,x} \end{Bmatrix}, \quad \epsilon_b^0 = \begin{Bmatrix} w_{b,xx} \\ w_{b,yy} \\ 2w_{b,xy} \end{Bmatrix}, \quad \epsilon_s^0 = \begin{Bmatrix} w_{s,x} \\ w_{s,y} \end{Bmatrix}, \quad \epsilon_{nl} = \frac{1}{2} \bar{A} \theta, \quad (16)$$

in which

$$\bar{A} = \begin{Bmatrix} (w_b + w_s)_x & 0 \\ 0 & (w_b + w_s)_y \\ (w_b + w_s)_y & (w_b + w_s)_x \end{Bmatrix}, \quad \theta = \begin{Bmatrix} (w_b + w_s)_x \\ (w_b + w_s)_y \end{Bmatrix}, \quad (17)$$

where ϵ_m^0 , ϵ_b^0 , ϵ_s^0 and ϵ_{nl} are the in-plane linear strain, bending strain, shear strain and nonlinear strain, respectively.

2.3. Constitutive relationship for PFGPs

For PFGPs, most of previous investigations for smart piezoelectric structures in thermal environment were carried out by using linear piezoelectric constitutive equation [26]

$$\begin{aligned} \sigma &= Q \tilde{\epsilon} - e^T E - Q \alpha \Delta T \\ D &= e \tilde{\epsilon} + g E + P \Delta T \\ S &= \alpha^T Q^T \tilde{\epsilon} + P^T E + \bar{C} \Delta T \end{aligned} \quad (18)$$

where σ and $\tilde{\epsilon}$ are the stress and strain vectors; Q is the elastic constant matrix; $\alpha = \{\alpha_x \quad \alpha_y \quad 0 \quad 0 \quad 0\}^T$ is the thermal expansion vector of piezoelectric materials; D is the electrical displacement vector; e is the piezoelectric stress constant matrix; g is the dielectric constant matrix; $P = [P_1 \quad P_2 \quad P_3]^T$ is the pyroelectrical vector; S and $\bar{C}$ are the entropy and specific heat, respectively; $\Delta T = T(z) - T_0$ is the difference between the temperature $T(z)$ and reference temperature T_0 .

And

$$Q = \begin{bmatrix} Q_m & \mathbf{0} \\ \mathbf{0} & Q_s \end{bmatrix}, \quad (19)$$

in which

$$Q_m = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}, \quad Q_s = \begin{bmatrix} Q_{44} & 0 \\ 0 & Q_{55} \end{bmatrix}, \quad (20)$$

with

$$\begin{aligned} Q_{11} &= Q_{22} = \frac{E(z)}{1 - \nu(z)^2}, \quad Q_{12} = Q_{21} = \frac{\nu(z)E(z)}{1 - \nu(z)^2}, \\ Q_{44} &= Q_{55} = \frac{E(z)}{2(1 + \nu(z))}. \end{aligned} \quad (21)$$

The forms of piezoelectric stress constant and dielectric constant matrices are

$$e = [e_m \quad e_s] = \begin{bmatrix} 0 & 0 & 0 & 0 & e_{15} \\ 0 & 0 & 0 & e_{24} & 0 \\ e_{31} & e_{32} & 0 & 0 & 0 \end{bmatrix}, \quad g = \begin{bmatrix} k_{11} & 0 & 0 \\ 0 & k_{22} & 0 \\ 0 & 0 & k_{33} \end{bmatrix}, \quad (22)$$

where

$$e_m = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ e_{31} & e_{32} & 0 \end{bmatrix}, \quad e_s = \begin{bmatrix} 0 & e_{24} \\ e_{15} & 0 \\ 0 & 0 \end{bmatrix}, \quad (23)$$

in which, the piezoelectric stress constants can be calculated by

$$\begin{Bmatrix} e_{31} \\ e_{32} \end{Bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{Bmatrix} d_{31} \\ d_{32} \end{Bmatrix}, \quad \begin{Bmatrix} e_{24} \\ e_{15} \end{Bmatrix} = \begin{bmatrix} Q_{44} & 0 \\ 0 & Q_{55} \end{bmatrix} \begin{Bmatrix} d_{24} \\ d_{15} \end{Bmatrix}, \quad (24)$$

where d_{31} , d_{32} , d_{15} and d_{24} are the piezoelectric strain constants.

In this work, an augmented piezoelectric constitutive equation [44] is introduced to study the nonlinear mechanical behavior closer to the reality. The piezoelectric stress and dielectric constant matrices in Eq. (18) are amended to be temperature-dependent with the forms as

$$\tilde{e} = e + \hat{e} \Delta T, \quad \tilde{g} = g + \hat{g} \Delta T, \quad (25)$$

where, e and g are the piezoelectric stress and dielectric constant matrices at a reference temperature T_0 ; $\hat{e}$ and $\hat{g}$ are the coefficients of piezoelectric stress and dielectric constants which need to be determined by experiments at different temperature.

Hence, the augmented form of the linear piezoelectric constitutive equation can be described as

$$\begin{aligned} \sigma &= Q \tilde{\epsilon} - \tilde{e}^T E - Q \alpha \Delta T \\ D &= \tilde{e} \tilde{\epsilon} + \tilde{g} E + P \Delta T \\ S &= \alpha^T Q^T \tilde{\epsilon} + P^T E + \bar{C} \Delta T \end{aligned} \quad (26)$$

The electric field vector E is thus calculated as

$$E = \{E_x \quad E_y \quad E_z\}^T = -\text{grad} \phi. \quad (27)$$

It should be noticed that only the electric field component in the z direction is considered in this study. Hence, $E = \{0 \quad 0 \quad E_z\}^T$. The stress resultants in PFGPs can be expressed as

$$(N_x, N_y, N_{xy}) = \int_{-h_t/2}^{h_t/2} (\sigma_x, \sigma_y, \sigma_{xy}) dz, \quad (28a)$$

$$(M_x, M_y, M_{xy}) = \int_{-h_t/2}^{h_t/2} (\sigma_x, \sigma_y, \sigma_{xy}) (z - z_0) dz, \quad (28b)$$

$$(\bar{Q}_x, \bar{Q}_y) = \int_{-h_t/2}^{h_t/2} (\sigma_{xz}, \sigma_{yz}) dz. \quad (28c)$$

Substituting Eq. (26) into Eq. (28), the stress resultants can be rewritten as

$$\begin{Bmatrix} N \\ M \\ \bar{Q} \end{Bmatrix} = \hat{c} \begin{Bmatrix} \epsilon_m^0 + \epsilon_{nl} \\ \epsilon_b^0 \\ \epsilon_s^0 \end{Bmatrix} - \begin{Bmatrix} N^p \\ M^p \\ \mathbf{0} \end{Bmatrix} - \begin{Bmatrix} N^{th} \\ M^{th} \\ \mathbf{0} \end{Bmatrix}, \quad (29)$$

with

$$\hat{c} = \begin{bmatrix} D_{mb} & \mathbf{0} \\ \mathbf{0} & C_s \end{bmatrix}, \quad N = \{N_x^T \quad N_y^T \quad N_{xy}^T\}^T, \quad M = \{M_x^T \quad M_y^T \quad M_{xy}^T\}^T \quad (30)$$

in which

$$D_{mb} = \begin{bmatrix} A & B \\ B & D \end{bmatrix}, \quad C_s = k \int_{-ht/2}^{ht/2} Q_{ij} dz, \quad i, j = 4, 5, \quad (31)$$

where k is the shear correction factor and $k = 5/6$ is used in this paper,

$$(A_{ij}, B_{ij}, D_{ij}) = \int_{-ht/2}^{ht/2} Q_{ij} (1, (z - z_0), (z - z_0)^2) dz, \quad i, j = 1, 2, 6, \quad (32)$$

due to the introduction of neutral surface, $B_{ij} = 0$, as in Ref. [45].

The stress resultants generated by the electric field are calculated by

$$N^p = \begin{Bmatrix} N_x^p \\ N_y^p \\ N_{xy}^p \end{Bmatrix} = \int_{\frac{h_f}{2}}^{\frac{h_f}{2}+h_p} \tilde{e}_m^T E dz + \int_{-\frac{h_f}{2}}^{-\frac{h_f}{2}-h_p} \tilde{e}_m^T E dz, \quad (33)$$

$$M^p = \begin{Bmatrix} M_x^p \\ M_y^p \\ M_{xy}^p \end{Bmatrix} = \int_{\frac{h_f}{2}}^{\frac{h_f}{2}+h_p} \tilde{e}_m^T E (z - z_0) dz + \int_{-\frac{h_f}{2}}^{-\frac{h_f}{2}-h_p} \tilde{e}_m^T E (z - z_0) dz. \quad (34)$$

The thermal stress are defined as

$$N^{th} = \begin{Bmatrix} N_x^{th} \\ N_y^{th} \\ N_{xy}^{th} \end{Bmatrix} = \int_{\frac{h_f}{2}}^{\frac{h_f}{2}+h_p} Q_{mf} \alpha_{mf} \Delta T dz + \int_{-\frac{h_f}{2}}^{-\frac{h_f}{2}-h_p} Q_{mp} \alpha_{mp} \Delta T dz + \int_{-\frac{h_f}{2}}^{\frac{h_f}{2}} Q_{mp} \alpha_{mp} \Delta T dz, \quad (35)$$

$$M^{th} = \begin{Bmatrix} M_x^{th} \\ M_y^{th} \\ M_{xy}^{th} \end{Bmatrix} = \int_{\frac{h_f}{2}}^{\frac{h_f}{2}+h_p} Q_{mf} \alpha_{mf} \Delta T (z - z_0) dz + \int_{-\frac{h_f}{2}}^{-\frac{h_f}{2}-h_p} Q_{mp} \alpha_{mp} \Delta T (z - z_0) dz + \int_{-\frac{h_f}{2}}^{\frac{h_f}{2}} Q_{mp} \alpha_{mp} \Delta T (z - z_0) dz. \quad (36)$$

In the above equations, the subscripts f , p represent the FGP and piezo-electric layers. $\alpha_m = \{\alpha_x \quad \alpha_y \quad 0\}^T$. Note that for the S2 structure, the integral ranges in stresses N^p , M^p , N^{th} and M^{th} are different with S1 structure.

3. Geometrically nonlinear isogeometric analysis based on S-FSDT

In this section, the displacements and electric potential of the plates are approximated by using the NURBS basis functions. Then, the isogeometric finite element motion equations are derived through Hamilton's variational principle.

3.1. Basis functions

Combining an additional individual weight ϑ_{ij} with the tensor product of B-spline basis functions in two parametric dimensions ξ and η with two knot vectors $\Xi(\xi) = \{\xi_1 = 0, \dots, \xi_i, \dots, \xi_{m+p+1} = 1\}$ and $\Xi(\eta) = \{\eta_1 = 0, \dots, \eta_j, \dots, \eta_{n+q+1} = 1\}$, the two dimensional (2D) NURBS basis functions can be written as [46]

$$R_{ij}^{p,q}(\xi, \eta) = \frac{G_{ip}(\xi) H_{jq}(\eta) \vartheta_{ij}}{\sum_{i=1}^m \sum_{j=1}^n G_{ip}(\xi) H_{jq}(\eta) \vartheta_{ij}}, \quad (37)$$

where G and H are the B-spline basis functions; i and j are the knot indexes; p and q are the orders; m and n are the number of basis functions.

3.2. Discrete equation for displacement field

The displacements in the neutral surface can be approximated as

$$u(\xi, \eta) = [u_0 \quad v_0 \quad w_b \quad w_s]^T = \sum_{k=1}^{m \times n} R_k(\xi, \eta) d_k, \quad (38)$$

where $d_k = [u_{0k} \quad v_{0k} \quad w_{bk} \quad w_{sk}]^T$ is the unknown displacement vector of control point k ; $R_k(\xi, \eta)$ is the NURBS basis function defined in Eq. (37).

The strain functions can be achieved by substituting Eq. (38) into Eq. (16):

$$\varepsilon_m^0 = \sum_{k=1}^{m \times n} B_k^m d_k, \quad \varepsilon_b^0 = \sum_{k=1}^{m \times n} B_k^b d_k, \quad \varepsilon_s^0 = \sum_{k=1}^{m \times n} B_k^s d_k, \quad \varepsilon_{nl} = \sum_{k=1}^{m \times n} \frac{1}{2} B_k^{nl} d_k, \quad (39)$$

where

$$B_k^m = \begin{bmatrix} R_{k,x} & 0 & 0 & 0 \\ 0 & R_{k,y} & 0 & 0 \\ R_{k,y} & R_{k,x} & 0 & 0 \end{bmatrix}, \quad (40a)$$

$$B_k^b = \begin{bmatrix} 0 & 0 & R_{k,xx} & 0 \\ 0 & 0 & R_{k,yy} & 0 \\ 0 & 0 & 2R_{k,xy} & 0 \end{bmatrix}, \quad (40b)$$

$$B_k^s = \begin{bmatrix} 0 & 0 & 0 & R_{k,x} \\ 0 & 0 & 0 & R_{k,y} \end{bmatrix}, \quad (40c)$$

$$B_k^{nl} = \bar{A} B_k^g, \quad (40d)$$

with

$$B_k^g = \begin{bmatrix} 0 & 0 & R_{k,xx} & R_{k,x} \\ 0 & 0 & R_{k,yy} & R_{k,y} \end{bmatrix}. \quad (41)$$

The generalized strain vector $\hat{\varepsilon}$ can be rewritten as

$$\hat{\varepsilon} = \begin{Bmatrix} \varepsilon_m^0 + \varepsilon_{nl} \\ \varepsilon_b^0 \\ \varepsilon_s^0 \end{Bmatrix} = \sum_{k=1}^{m \times n} \left(B_k^L + \frac{1}{2} B_k^{NL} \right) d_k = \sum_{k=1}^{m \times n} \left(\begin{Bmatrix} B_k^m \\ B_k^b \\ B_k^s \end{Bmatrix} + \frac{1}{2} \begin{Bmatrix} B_k^{nl} \\ 0 \\ 0 \end{Bmatrix} \right) d_k, \quad (42)$$

in which

$$B_k^L = \left\{ (B_k^m)^T \quad (B_k^b)^T \quad (B_k^s)^T \right\}^T, \quad B_k^{NL} = \left\{ (B_k^{nl})^T \quad 0 \quad 0 \right\}^T. \quad (43)$$

The variation of the strain is defined as

$$\delta \hat{\varepsilon} = \sum_{k=1}^{m \times n} (B_k^L + B_k^{NL}) d_k = \sum_{k=1}^{m \times n} \left(\begin{Bmatrix} B_k^m \\ B_k^b \\ B_k^s \end{Bmatrix} + \begin{Bmatrix} B_k^{nl} \\ 0 \\ 0 \end{Bmatrix} \right) d_k. \quad (44)$$

3.3. Discrete equation for electric field

It is assumed that the electric potential vary linearly along the thickness [47]. The electric potential can be expressed as

$$\varphi = \sum_{k=1}^{m \times n} R_k \varphi_k \quad (45)$$

Substituting Eq. (45) into Eq. (27), the electric field can be written as

$$E = -\nabla \varphi = - \sum_{k=1}^{m \times n} B_{\phi k} \varphi_k. \quad (46)$$

where

$$B_{\phi k} = \{0 \quad 0 \quad R_k/h_p\}^T. \quad (47)$$

3.4. Isogeometric finite element motion equations

The Galerkin weak form of isogeometric finite element motion equations for PFGPs can be obtained by using Hamilton's principle denoted as

Table 1

Comparison of the lowest ten natural frequencies (Hz) of a SSSS PFGP between the developed method and exact solutions [49].

Mode	Method	<i>n</i>						
		0	0.5	1	5	15	100	1000
1	Present	145.355	186.457	200.520	232.386	249.279	261.317	263.695
Exact [49]	145.350	186.254	200.569	233.036	250.327	262.674	265.114	
2/3	Present	363.061	465.746	500.893	580.510	622.691	652.743	658.678
Exact [49]	363.050	465.223	500.994	582.112	625.288	656.108	662.201	
4	Present	580.371	744.551	800.771	928.076	995.549	1043.499	1052.981
Exact [49]	580.350	743.699	800.911	930.617	999.616	1048.857	1058.589	
5/6	Present	725.073	930.217	1000.481	1159.557	1243.759	1303.721	1315.561
Exact [49]	725.000	929.078	1000.571	1162.638	1248.819	1310.310	1322.461	
7/8	Present	941.718	1208.213	1299.527	1506.189	1615.520	1693.356	1708.723
Exact [49]	941.640	1206.740	1299.640	1510.197	1622.102	1701.923	1717.694	
9/10	Present	1230.273	1578.522	1697.911	1967.995	2110.774	2212.387	2232.444
Exact [49]	1229.880	1576.201	1697.617	1972.734	2118.843	2223.021	2243.603	

Table 2

Temperature-dependent material parameters of FGMS.

Material	Property	P_0	P_{-1}	P_1	P_2	P_3
Si ₃ N ₄	E (Pa)	348.430×10^9	0	-3.070×10^{-4}	2.160×10^{-7}	-8.946×10^{-11}
	α_{fx}, α_{fy}	5.8723×10^{-6}	0	9.095×10^{-4}	0	0
SUS304	E (Pa)	201.040×10^9	0	3.079×10^{-4}	-6.534×10^{-7}	0
	α_{fx}, α_{fy}	12.330×10^{-6}	0	8.086×10^{-4}	0	0

Table 3

Material parameters of PZT-4.

E_{11} (GPa)	81.3	ν_{23}	0.432
E_{22} (GPa)	81.3	ρ (kg/m ³)	7600
E_{33} (GPa)	64.5	e_{31} (C/m ²)	-5.2
G_{12} (GPa)	30.6	e_{32} (C/m ²)	-5.2
G_{31} (GPa)	25.6	k_{33} (C ² /(Nm ²))	11505.0×10^{-12}
G_{23} (GPa)	25.6	α_p (°C)	1.2×10^{-6}
ν_{12}	0.329	P_3 (C/m ² °C)	0.25×10^{-4}
ν_{13}	0.432	k_p (W/mK)	2.1

$$\delta \int_{t_1}^{t_2} L dt = \delta \int_{t_1}^{t_2} (T - U + W_{ext}) dt = 0, \quad (48)$$

where L is the total energy including kinetic energy T , potential energy U , and external work W_{ext} .

The elementary kinetic energy, potential energy and external work are expressed as

$$T^e = \int_{V^e} \frac{1}{2} \dot{\mathbf{m}} \dot{\mathbf{u}}^T \dot{\mathbf{u}} dV^e, \quad (49a)$$

$$U^e = \int_{V^e} \left(\frac{1}{2} \tilde{\mathbf{e}}^T \mathbf{Q} \tilde{\mathbf{e}} - \tilde{\mathbf{e}}^T \tilde{\mathbf{e}}^T \mathbf{E} - \frac{1}{2} \mathbf{E}^T \tilde{\mathbf{g}} \mathbf{E} - \tilde{\mathbf{e}}^T \mathbf{Q} \alpha \Delta T - \mathbf{E} P \Delta T - \frac{1}{2} \tilde{\mathbf{C}} \Delta T^2 \right) dV^e, \quad (49b)$$

$$W_{ext}^e = \int_{\Omega^e} (\mathbf{u}^T \mathbf{f}_s - \varphi \mathbf{q}_s) d\Omega^e + \sum \mathbf{u}^T \mathbf{f}_p + \int_{V^e} \mathbf{u}^T \mathbf{f}_b dV^e, \quad (49c)$$

where $\mathbf{u}$ is the mechanical displacement; $\dot{\mathbf{u}}$ is velocity; $\mathbf{q}_s$ denotes the surface charge; $\mathbf{f}_p$, $\mathbf{f}_s$ and $\mathbf{f}_b$ are the mechanical point loads, surface loads, and body loads, respectively.

By substituting Eqs. (38), (42), (44), (45) and (46) into Eq. (48) and assembling elements, the nonlinear governing equations of PFGPs are denoted as

$$\mathbf{M}_{uu} \ddot{\mathbf{d}} + (\mathbf{K}_{uu}^L + \mathbf{K}_{uu}^{NL} - \mathbf{K}_0) \mathbf{d} + \mathbf{K}_{u\varphi} \boldsymbol{\varphi} = \mathbf{f} + \mathbf{f}_{th}, \quad (50a)$$

$$\mathbf{K}_{\varphi u} \mathbf{d} - \mathbf{K}_{\varphi\varphi} \boldsymbol{\varphi} = \mathbf{f}_{\varphi} - \mathbf{f}_{\varphi\theta}. \quad (50b)$$

where $\mathbf{K}_{uu}^L$ and $\mathbf{K}_{uu}^{NL}$ are the linear, nonlinear stiffness matrices, respectively. And,

$$\mathbf{M}_{uu} = \int_V \bar{\mathbf{A}}^T \mathbf{m} \bar{\mathbf{A}} dV, \quad \mathbf{K}_0 = \mathbf{K}_0^p + \mathbf{K}_0^{th}, \quad \mathbf{K}_{u\varphi} = \int_V \mathbf{B}^L \mathbf{e}^{*T} \mathbf{B}_{\varphi} dV, \quad (51)$$

$$\mathbf{K}_{uu}^{NL} = \int_V \left(\frac{1}{2} (\mathbf{B}^L)^T \tilde{\mathbf{c}} \mathbf{B}^{NL} + (\mathbf{B}^{NL})^T \tilde{\mathbf{c}} \mathbf{B}^L + \frac{1}{2} (\mathbf{B}^{NL})^T \tilde{\mathbf{c}} \mathbf{B}^{NL} \right) dV, \quad (52)$$

$$\mathbf{K}_{uu}^L = \int_V (\mathbf{B}^L)^T \tilde{\mathbf{c}} \mathbf{B}^L dV = \int_V \left(\left\{ \begin{matrix} \mathbf{B}^m \\ \mathbf{B}^b \end{matrix} \right\}^T \mathbf{D}_{mb} \left\{ \begin{matrix} \mathbf{B}^m \\ \mathbf{B}^b \end{matrix} \right\} + (\mathbf{B}^s)^T \mathbf{C}_s \mathbf{B}^s \right) dV, \quad (53)$$

$$\mathbf{K}_{\varphi u} = \mathbf{K}_{u\varphi}^T, \quad \mathbf{K}_{\varphi\varphi} = \int_V \mathbf{B}_{\varphi}^T \tilde{\mathbf{g}} \mathbf{B}_{\varphi} dV, \quad (54)$$

$$\mathbf{f} = \int_{\Omega} \mathbf{R}^T \mathbf{f}_s d\Omega + \sum \mathbf{R}^T \mathbf{f}_p + \int_V \mathbf{R}^T \mathbf{f}_b dV, \quad (55)$$

$$\mathbf{f}_{th} = \int_V \left((\mathbf{B}^m)^T \mathbf{N}^{th} + (\mathbf{B}^b)^T \mathbf{M}^{th} \right) dV,$$

$$\mathbf{f}_{\varphi} = \int_{\Omega} \mathbf{R}^T \mathbf{q}_E d\Omega, \quad \mathbf{q}_E = \tilde{k}_{33} \frac{U_0}{h_p}, \quad \mathbf{f}_{\varphi\theta} = \int_V \mathbf{B}_{\varphi}^T \mathbf{P} \Delta T dV, \quad (56)$$

where U_0 is the voltage. And

$$\mathbf{e}^* = [\tilde{\mathbf{e}}_m^T \quad -(\mathbf{z} - \mathbf{z}_0) \tilde{\mathbf{e}}_m^T \quad \tilde{\mathbf{e}}_s^T]^T, \quad (57)$$

$$\mathbf{m} = \begin{bmatrix} I_0 & I_1 \\ I_1 & I_2 \end{bmatrix}, \quad (58)$$

$$\bar{\mathbf{A}} = [\mathbf{A}_1 \quad \mathbf{A}_2]^T, \quad (59)$$

with

$$(I_0, I_1, I_2) = \int_{-h_t/2}^{h_t/2} \rho(\mathbf{z}) \left(1, (\mathbf{z} - \mathbf{z}_0), (\mathbf{z} - \mathbf{z}_0)^2 \right) d\mathbf{z}, \quad (60)$$

$$\mathbf{A}_1 = \begin{bmatrix} R_k & 0 & 0 & 0 \\ 0 & R_k & 0 & 0 \\ 0 & 0 & R_k & R_k \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 0 & 0 & R_{k,x} & 0 \\ 0 & 0 & R_{k,y} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (61)$$

$\mathbf{K}_0^p$ and $\mathbf{K}_0^{th}$ are the initial stress stiffness matrices caused by electrical and thermal loads, which can be expressed as

$$\mathbf{K}_0^p = \int_V (\mathbf{B}^g)^T \begin{bmatrix} N_x^p & N_{xy}^p \\ N_{xy}^p & N_y^p \end{bmatrix} \mathbf{B}^g dV, \quad \mathbf{K}_0^{th} = \int_V (\mathbf{B}^g)^T \begin{bmatrix} N_x^{th} & N_{xy}^{th} \\ N_{xy}^{th} & N_y^{th} \end{bmatrix} \mathbf{B}^g dV, \quad (62)$$

Table 4

Comparison of the dimensionless natural frequencies of a CCCC PFGP between the developed method and reference solutions using FEM based on HSDT ($V_u = V_l = 0$ V) [26].

Thermal			$\bar{\omega}_{NL}$										
			$\bar{W}_{\max}/h_f$										
condition	n	$\bar{\omega}_L$	0.2		0.4		0.6		0.8		1		
$T_u = 300$ K, $T_l = 300$ K		Present	[26]	Present	[26]	Present	[26]	Present	[26]	Present	[26]	Present	[26]
	0	20.74	20.73	21.02	21.00	21.84	21.76	23.14	22.97	24.83	24.55	26.84	26.43
	0.2	17.21	17.24	17.44	17.46	18.13	18.11	19.21	19.13	20.62	20.41	22.29	22.04
	1	13.10	13.10	13.28	13.26	13.79	13.77	14.61	14.50	15.68	15.49	16.95	16.66
	5	10.78	10.74	10.92	10.86	11.34	11.25	11.99	11.86	12.84	12.66	13.85	13.60
$T_u = 500$ K, $T_l = 300$ K	∞	9.55	9.55	9.68	9.67	10.06	10.01	10.65	10.57	11.42	11.28	12.35	12.14
	0	20.28	20.27	20.56	20.53	21.39	21.30	22.67	22.51	24.38	24.10	26.39	25.97
	0.2	16.81	16.83	17.04	17.06	17.73	17.71	18.82	18.71	20.23	20.03	21.91	21.57
	1	12.76	12.76	12.94	12.94	13.46	13.41	14.29	14.18	15.37	15.18	16.64	16.36
	5	10.47	10.42	10.61	10.55	11.03	10.93	11.69	11.56	12.55	12.32	13.57	13.26
$T_u = 300$ K, $T_l = 100$ K	∞	9.24	9.23	9.37	9.35	9.75	9.70	10.36	10.27	11.14	10.99	12.07	11.86
	0	21.20	21.19	21.48	21.46	22.30	22.23	23.60	23.43	25.30	25.02	27.32	26.90
	0.2	17.58	17.62	17.82	17.84	18.50	18.49	19.58	19.51	20.98	20.79	22.66	22.38
	1	13.36	13.36	13.54	13.54	14.05	14.00	14.86	14.80	15.92	15.73	17.18	16.91
	5	11.00	10.95	11.14	11.08	11.54	11.46	12.18	12.06	13.02	12.78	14.02	13.77
	∞	9.75	9.74	9.88	9.86	10.25	10.20	10.83	10.74	11.59	11.45	12.50	12.29

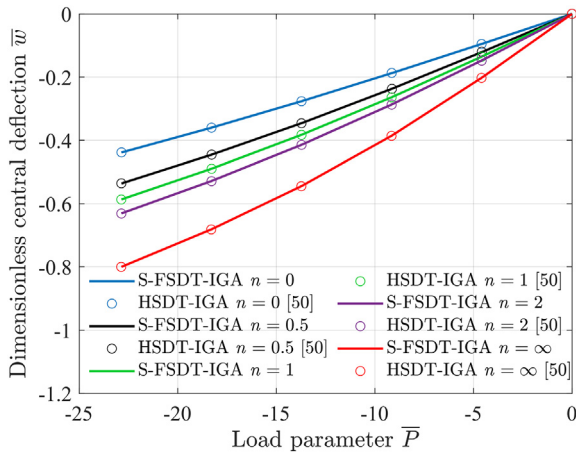


Fig. 2. Variation of the dimensionless nonlinear central deflections with the mechanical load parameter. The present results match well the reference solutions using HSDT [50].

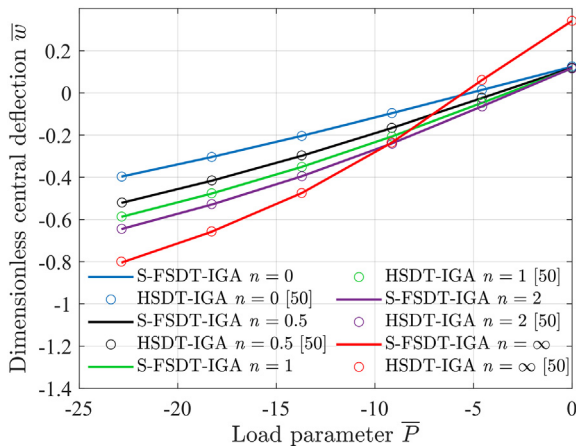


Fig. 3. Variation of the dimensionless nonlinear central deflections with the mechanical load parameter in thermal environment. The present results match well the reference solutions using HSDT [50].

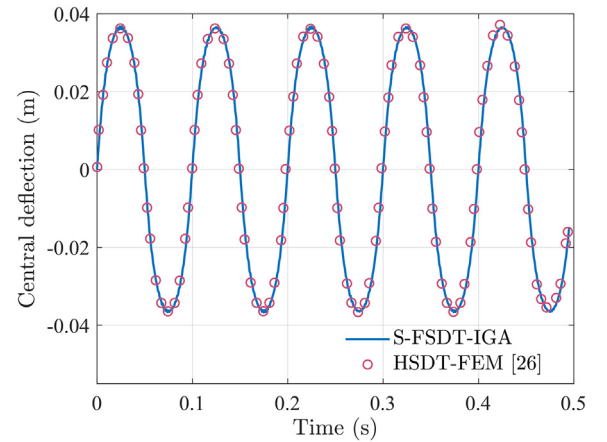


Fig. 4. Central deflection of the plate under electrical load $V_u = V_l = 200$ V and temperature rise $T_u = 600$ K, $T_l = 300$ K. The present approach offers remarkable results as compared with the existing solutions [26].

Eliminating the potential φ in Eq. (50), the governing equations are rewritten as

$$M_{uu}\ddot{d} + \bar{K}d = F. \quad (63)$$

Therefore, the free vibration and static bending response can be solved by

$$(\bar{K} - \omega_n^2 M_{uu})d = 0, \quad (64)$$

$$\bar{K}d = F, \quad (65)$$

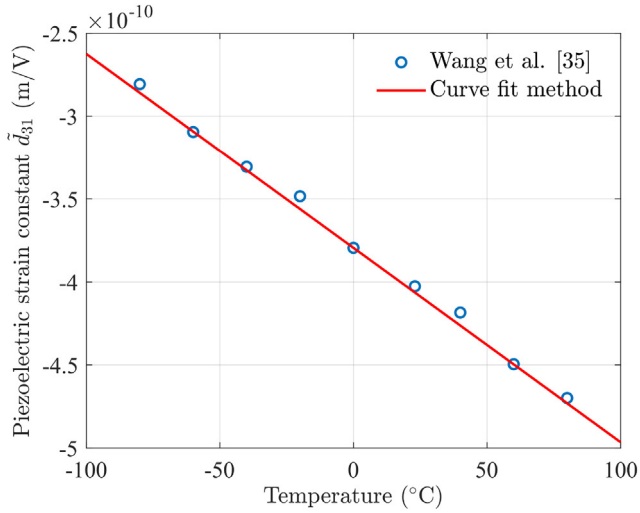
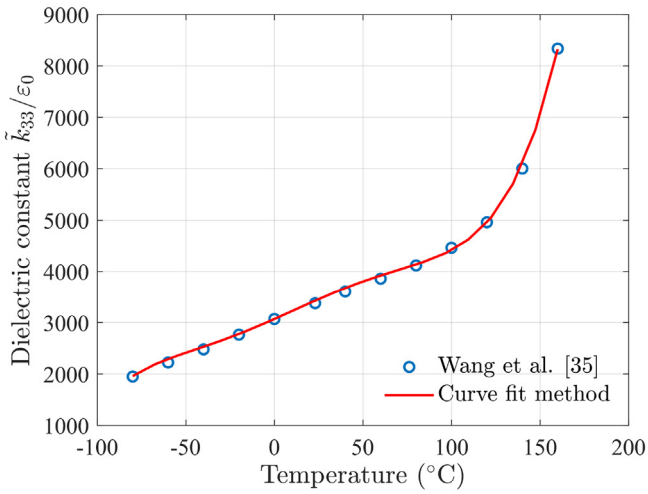
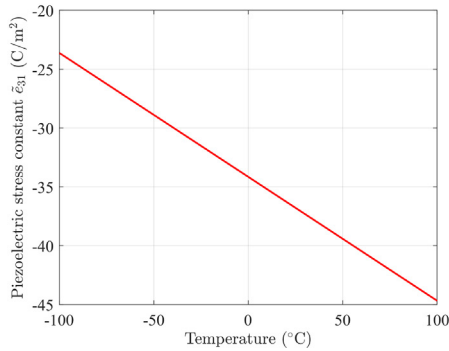
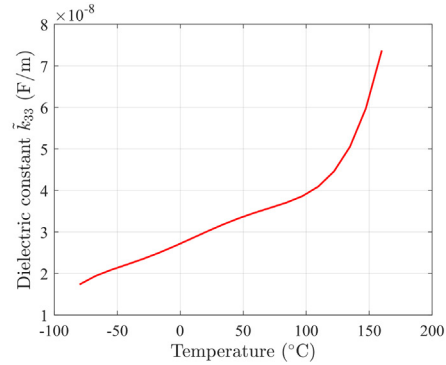
where

$$\bar{K} = K_{uu}^L + K_{uu}^{NL} - K_0 + K_{up}K_{\varphi\varphi}^{-1}K_{\varphi u}, \quad (66)$$

$$F = f + f_{th} + K_{up}K_{\varphi\varphi}^{-1}f_{\varphi} - K_{up}K_{\varphi\varphi}^{-1}f_{\varphi\varphi}. \quad (67)$$

4. Solution procedure

The geometrically nonlinear problems of PFGPs considered in this work can be classified into three types: nonlinear natural frequencies, nonlinear static bending and nonlinear dynamic analyses.

(a) $\tilde{d}_{31}$ (b) $\tilde{k}_{33}/\epsilon_0$ Fig. 5. Trend of $\tilde{d}_{31}$ and $\tilde{k}_{33}/\epsilon_0$ with temperature.(a) $\tilde{e}_{31}$ (b) $\tilde{k}_{33}$ Fig. 6. Trend of $\tilde{e}_{31}$ and $\tilde{k}_{33}$ with temperature.

4.1. Nonlinear natural frequencies

The technique presented by Sundararajan et al. [9] is introduced to solve the nonlinear natural frequencies of PFGPs. In this setting, the eigenvector is calculated first by neglecting the nonlinear stiffness matrix K_{uu}^{NL} in Eq. (64); next, scale up the eigenvector so that the maximum displacement is equal to the given amplitude ($\bar{W}_{\max}/h_f$) for obtaining K_{uu}^{NL} ; then a new eigenvalue and eigenvector can be solved through Eq. (64); this new eigenvector is amplified to the same amplitude for calculating a new K_{uu}^{NL} ; the iterative process is executed repeatedly until the error norm of the natural frequencies are within the tolerance.

$$\sqrt{\frac{(\omega^{r+1} - \omega^r)^2}{(\omega^{r+1})^2}} \leq \text{tolerance}, \quad (68)$$

where ω and r represent the frequency and iteration number; the tolerance is set to be 1×10^{-3} in this paper.

4.2. Nonlinear static bending analysis

In terms of nonlinear bending analysis, a residual force Ψ is introduced in Eq. (65), then the nonlinear discrete system equation is rewritten as

$$\Psi = \bar{K}d - F. \quad (69)$$

The Newton–Raphson method [48] is used for solving the above equation. In this iterative process, the displacements vector ^{i+1}d at the $(i+1)$ th iteration can be obtained according to the solution at the previous iteration ($^i d$), that is

$$^{i+1}d = ^i d + ^i \Delta d, \quad (70)$$

in which $^i \Delta d$ is the incremental displacement and given by

$$^i \Delta d = -^i \Psi / K_T = \left[F - \left(K_{uu}^L + ^i K_{uu}^{NL} (^i d) - K_0 + K_{u\varphi} K_{\varphi\varphi}^{-1} K_{\varphi u} \right) ^i d \right] / K_T, \quad (71)$$

where K_T is the tangent stiffness matrix and defined as

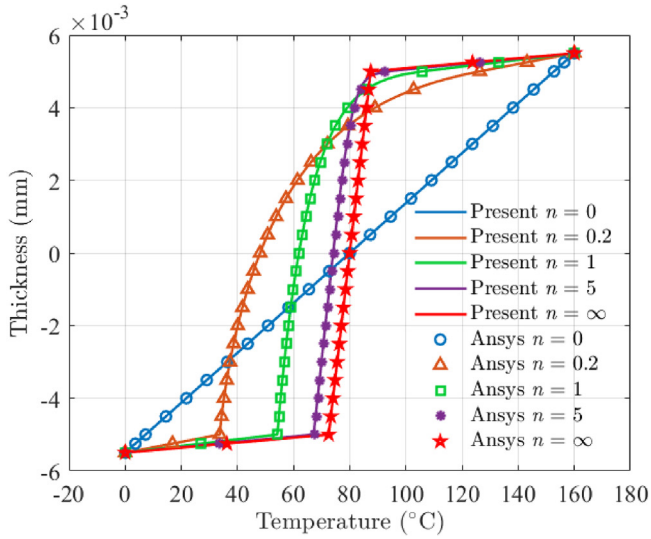
$$K_T = \partial \Psi (^i d) / \partial ^i d = \bar{K}^{NL} + K_G, \quad (72)$$

in which

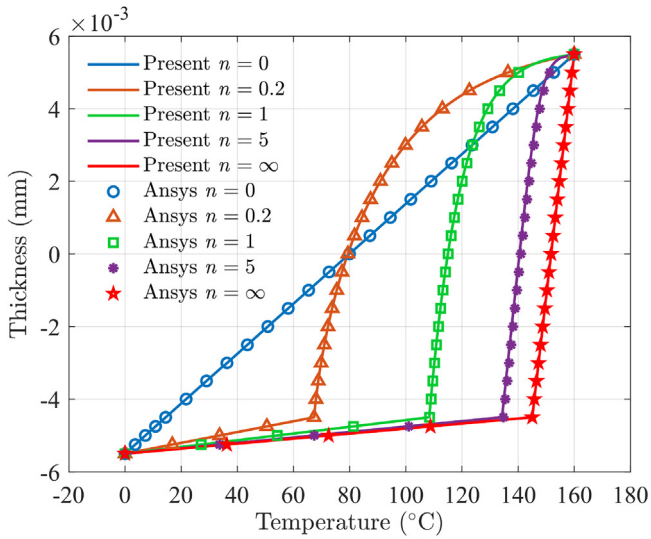
$$\bar{K}^{NL} = \int_V (B^L + B^{NL})^T \hat{c} (B^L + B^{NL}) dV. \quad (73)$$

The geometric stiffness matrix K_G is expressed as

$$K_G = \int_{\Omega} (B^g)^T \bar{N} \bar{B}^g d\Omega, \quad (74)$$



(a) S1 structure



(b) S2 structure

Fig. 7. Temperature filed through the thickness of S1 and S2 structures. The present method yields acceptable results compared with the FEM using ANSYS.

in which $\bar{N} = \begin{bmatrix} N_x & N_{xy} \\ N_{xy} & N_y \end{bmatrix}$ and can be confirmed by in-plane forces in Eq. (29).

The specific iterative process is briefly described as follows:

- (1) Solve Eq. (65) for the linear displacement by neglecting the nonlinear stiffness matrix;
- (2) Take the linear solution as an approximate solution for the first iteration 1d ;
- (3) Solve $^1\Psi$ by substituting 1d into Eq. (69);
- (4) Solve K_T according to Eq. (72);
- (5) Solve $^1\Delta d$ through Eq. (71);
- (6) Calculate 2d through Eq. (70);
- (7) Repeat step (3)–(6) until the error norm of displacement satisfies the convergence criteria as

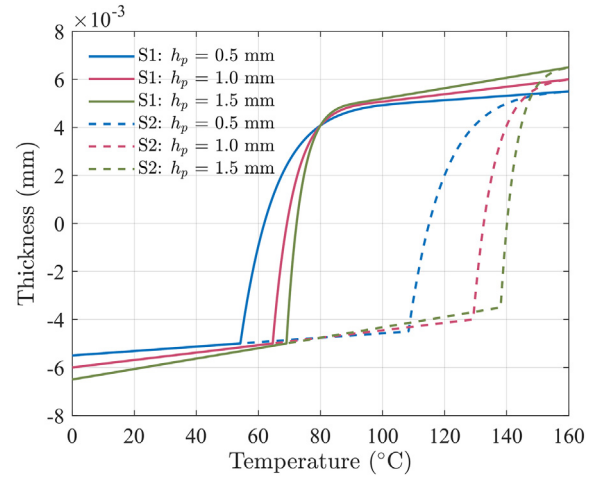


Fig. 8. Temperature distribution of S1 and S2 structures with different thicknesses of piezoelectric layers.

$$\text{error} = \frac{\|^{i+1}d - ^id\|}{\|^{i+1}d\|} \leq \text{tolerance}. \quad (75)$$

4.3. Nonlinear transient dynamic analysis

The Newmark- β direct integration method in combination with Picard method [48] is implemented to discretize the nonlinear dynamic problem denoted in Eq. (63). After the application of Newmark's integration, Eq. (63) can be described as

$$\hat{K}_{m+1}d_{m+1} = \hat{F}_{m+1}, \quad (76)$$

where $\hat{K}_{m+1}$ and $\hat{F}_{m+1}$ denote the effective stiffness matrix and force vector expressed as

$$\hat{K}_{m+1} = \bar{K}_{m+1} + a_0 M_{uu}, \quad (77)$$

$$\hat{F}_{m+1} = F_{m+1} + M_{uu} \left(a_0 d_m + a_1 \dot{d}_m + a_2 \ddot{d}_m \right), \quad (78)$$

where

$$a_0 = \frac{1}{\beta \Delta t^2}, \quad a_1 = \frac{1}{\beta \Delta t}, \quad a_2 = \frac{1}{2\beta} - 1. \quad (79)$$

Once the displacement d_{m+1} at time $t = (m+1)\Delta t$ is known, the acceleration and velocity at time $t = (m+1)\Delta t$ can be computed from

$$\ddot{d}_{m+1} = a_0(d_{m+1} - d_m) - a_1 \dot{d}_m - a_2 \ddot{d}_m, \quad (80)$$

$$\dot{d}_{m+1} = \dot{d}_m + a_3 \ddot{d}_m - a_4 \ddot{d}_{m+1}, \quad (81)$$

where

$$a_3 = (1 - \gamma)\Delta t, \quad a_4 = \gamma\Delta t. \quad (82)$$

It can be seen the effective stiffness matrix depends on the unknown displacement at time $t = (m+1)\Delta t$ because the existence of K_{uu}^{NL} in Eq. (66). Hence, the nonlinear system equation (Eq. (76)) need to be solved by using Picard algorithm as

$$^{i+1}\hat{K}(^id_{m+1})^{i+1}d_{m+1} = \hat{F}_{m+1}. \quad (83)$$

We can see that the effective stiffness matrix of $(i+1)$ th iteration is computed by using the displacement vector of i th. Such computation are repeated till the convergence criteria in Eq. (75) is satisfied.

Table 5The dimensionless natural frequencies of S1 plate structure ($V_u = V_l = 0$ V).

Thermal conditions	n	$\bar{\omega}_L$	$\bar{\omega}_{NL}$				
			$\bar{W}_{max}/h_f$				
			0.2	0.4	0.6	0.8	1.0
Temperature-independent piezoelectric coefficients (e_{31}, k_{33}) are used							
$T_u = 160$ °C, $T_l = 0$ °C	0	13.3687	13.5043	13.9023	14.5393	15.3834	16.4009
	0.2	12.7708	12.9024	13.2885	13.9061	14.7238	15.7086
	1	11.6649	11.7860	12.1409	12.7081	13.4587	14.3623
	5	10.9768	11.0802	11.3838	11.8705	12.5172	13.2989
	100	10.0996	10.1976	10.4855	10.9466	11.5584	12.2967
	∞	9.9794	10.0775	10.3658	10.8274	11.4395	12.1776
$T_u = 0$ °C, $T_l = 0$ °C	0	13.9277	14.0572	14.4379	15.0492	15.8626	16.8474
	0.2	13.3483	13.4735	13.8415	14.4323	15.2180	16.1688
	1	12.3491	12.4626	12.7962	13.3318	14.0448	14.9082
	5	11.7445	11.8403	12.1222	12.5766	13.1841	13.9234
	100	10.9382	11.0278	11.2916	11.7168	12.2852	12.9768
	∞	10.8276	10.9171	11.1808	11.6057	12.1736	12.8641
Temperature-dependent piezoelectric coefficients ($\bar{e}_{31}, \bar{k}_{33}$) are used							
$T_u = 160$ °C, $T_l = 0$ °C	0	13.3689	13.5055	13.9034	14.5401	15.3839	16.4013
	0.2	12.7719	12.9036	13.2897	13.9073	14.7250	15.7099
	1	11.6863	11.8073	12.1618	12.7285	13.4785	14.3814
	5	11.0099	11.1132	11.4162	11.9023	12.5481	13.3289
	100	10.1406	10.2383	10.5255	10.9857	11.5963	12.3334
	∞	10.0212	10.1191	10.4067	10.8673	11.4782	12.2151

Table 6The dimensionless natural frequencies of S2 plate structure ($V_u = V_l = 0$ V).

Thermal conditions	n	$\bar{\omega}_L$	$\bar{\omega}_{NL}$				
			$\bar{W}_{max}/h_f$				
			0.2	0.4	0.6	0.8	1.0
Temperature-independent piezoelectric coefficients (e_{31}, k_{33}) are used							
$T_u = 160$ °C, $T_l = 0$ °C	0	13.3554	13.4937	13.8948	14.5354	15.3835	16.4051
	0.2	12.6325	12.7687	13.1630	13.7914	14.6217	15.6199
	1	11.4738	11.6010	11.9676	12.5509	13.3202	14.2435
	5	10.5729	10.6849	11.0066	11.5188	12.1955	13.0091
	100	9.2321	9.3441	9.6647	10.1723	10.8385	11.6339
	∞	9.0564	9.1694	9.4926	10.0038	10.6737	11.4724
$T_u = 0$ °C, $T_l = 0$ °C	0	13.9646	14.0961	14.4784	15.0910	15.9056	16.8916
	0.2	13.4834	13.6100	13.9775	14.5663	15.3493	16.2973
	1	12.6680	12.7818	13.1113	13.6395	14.3431	15.1965
	5	12.0194	12.1166	12.3965	12.8464	13.4481	14.1810
	100	10.9083	11.0015	11.2694	11.6992	12.2728	12.9700
	∞	10.7628	10.8562	11.1247	11.5554	12.1296	12.8272
Temperature-dependent piezoelectric coefficients ($\bar{e}_{31}, \bar{k}_{33}$) are used							
$T_u = 160$ °C, $T_l = 0$ °C	0	13.3589	13.4971	13.8983	14.5388	15.3868	16.4084
	0.2	12.6563	12.7924	13.1864	13.8145	14.6442	15.6418
	1	11.5303	11.6573	12.0232	12.6052	13.3731	14.2950
	5	10.6371	10.7490	11.0700	11.5810	12.2563	13.0685
	100	9.2879	9.3999	9.7198	10.2262	10.8910	11.6851
	∞	9.1117	9.2247	9.5471	10.0571	10.7256	11.5230

5. Numerical examples and discussion

In this section, the IGA integrated with the S-FSDT is applied to solve several examples, which is to show its accuracy and performance. Comparison on the calculated results with respect to reference solutions is presented and discussed. Then, considering the influence of temperature on the piezoelectric stress and dielectric coefficients, we investigate the nonlinear natural frequencies, static bending and dynamic responses of two types of PFGPs as shown in Fig. 1. For convenience, the simply supported, clamped and free edges are simplified as S, C, and F respectively.

5.1. Comparison studies

In the comparison studies, S1 plate structure are employed to manifest the accuracy and effectiveness of the solution procedure for nonlinear free vibration, static bending and dynamic analysis. The linear piezoelectric constitutive equation Eq. (18) is still adopted here because the augmented equation Eq. (26) need to be established based on experimental data. The Young's modulus and thermal expansion coefficient of the FGMs are assumed temperature-dependent while the remaining properties of FGMs and piezoelectric materials are assumed to be temperature-independent.

Table 7The dimensionless natural frequencies of S1 plate structure ($T_u = 160\text{ }^\circ\text{C}$, $T_l = 0\text{ }^\circ\text{C}$).

Voltage	n	$\bar{\omega}_L$	$\bar{\omega}_{NL}$				
			$\bar{W}_{max}/h_f$				
			0.2	0.4	0.6	0.8	1.0
$V_u = 200\text{ V}$, $V_l = 200\text{ V}$	0	13.3420	13.4779	13.8765	14.5142	15.3591	16.3775
	0.2	12.7779	12.9095	13.2954	13.9127	14.7301	15.7146
	1	11.6931	11.8140	12.1682	12.7346	13.4842	14.3867
	5	11.0173	11.1204	11.4233	11.9090	12.5544	13.3348
	100	10.1486	10.2463	10.5332	10.993	11.6033	12.3399
$V_u = -200\text{ V}$, $V_l = -200\text{ V}$	∞	10.0294	10.1272	10.4145	10.8748	11.4852	12.2216
	0	13.3307	13.4667	13.8656	14.5039	15.3495	16.3686
	0.2	12.7659	12.8976	13.2840	13.9018	14.7199	15.7051
	1	11.6796	11.8007	12.1554	12.7224	13.4728	14.3761
	5	11.0026	11.1059	11.4092	11.8956	12.5418	13.3230
	100	10.1325	10.2303	10.5177	10.9783	11.5894	12.3269
	∞	10.0130	10.1110	10.3989	10.8598	11.4712	12.2085

Table 8The dimensionless natural frequencies of S2 plate structure ($T_u = 160\text{ }^\circ\text{C}$, $T_l = 0\text{ }^\circ\text{C}$).

Voltage	n	$\bar{\omega}_L$	$\bar{\omega}_{NL}$				
			$\bar{W}_{max}/h_f$				
			0.2	0.4	0.6	0.8	1.0
$V_u = 200\text{ V}$, $V_l = 200\text{ V}$	0	13.3635	13.5017	13.9027	14.5430	15.3907	16.4120
	0.2	12.6616	12.7977	13.1916	13.8193	14.6487	15.6460
	1	11.5367	11.6637	12.0293	12.6110	13.3785	14.3000
	5	10.6444	10.7563	11.0771	11.5877	12.2626	13.0744
	100	9.2967	9.4085	9.7281	10.2340	10.8983	11.6918
$V_u = -200\text{ V}$, $V_l = -200\text{ V}$	∞	9.1206	9.2335	9.5556	10.0651	10.7330	11.5298
	0	13.3542	13.4925	13.8938	14.5346	15.3828	16.4047
	0.2	12.6509	12.7871	13.1813	13.8096	14.6397	15.6376
	1	11.5239	11.6510	12.0171	12.5995	13.3677	14.2899
	5	10.6297	10.7417	11.0629	11.5743	12.2500	13.0627
	100	9.2792	9.3913	9.7114	10.2183	10.8837	11.6783
	∞	9.1028	9.2159	9.5386	10.0491	10.7181	11.5161

5.1.1. Example #1: Natural frequency

The linear natural frequencies of a fully simply supported (SSSS) square FGP without consideration of the effect of temperature is investigated. The dimensions of the plate are: $a = b = 400\text{ mm}$, $h_f = 5\text{ mm}$, $h_p = 0.1\text{ mm}$. The material of two piezoelectric layers are PZT-G1195N. The FGP consists of aluminum oxide and Ti-6Al-4 V. The material properties adopted are: for Aluminum oxide, $E = 320.24\text{ GPa}$, $\nu = 0.26$, $\rho = 3750\text{ kg/m}^3$; for Ti-6Al-4 V, $E = 105.70\text{ GPa}$, $\nu = 0.2981$, $\rho = 4429\text{ kg/m}^3$; and for PZT-G1195N, $E = 63.0\text{ GPa}$, $\nu = 0.30$, $\rho = 7600\text{ kg/m}^3$, $d_{31} = d_{32} = 2.54 \times 10^{-10}\text{ m/V}$, $k_{33} = 15.0 \times 10^{-9}\text{ F/m}$.

The lowest ten natural frequencies of the plate are computed by the developed method and the obtained results are then given in Table 1. In comparison with analytical solutions reported in [49], the developed method yields acceptable natural frequencies for different values of the gradient index n . This comparison essentially confirms the high accuracy of the present approach. The gradient index n has a significant impact on the natural frequencies, which increase with increasing the index.

5.1.2. Example #2: Nonlinear natural frequency

To investigate the linear and nonlinear natural frequencies, we consider a fully clamped (CCCC) $\text{Si}_3\text{N}_4/\text{SUS304}$ FGP with surface-bonded PZT-4 piezoelectric layers. The thickness of the FGP and each piezoelectric layer are 40 mm and 1 mm. The length and width of the plate are 400 mm. The temperature-dependent material properties of $\text{Si}_3\text{N}_4/\text{SUS304}$ plate and the material properties of piezoelectric layers are listed in Tables 2 and 3, respectively. The temperature-independent material properties of FGMs are: for Si_3N_4 , $\nu = 0.28$, $\rho =$

2370 kg/m^3 , $k_f = 9.19\text{ W/mK}$; and for SUS304, $\nu = 0.28$, $\rho = 8166\text{ kg/m}^3$, $k_f = 12.04\text{ W/mK}$.

Ignoring the applied voltages ($V_u = V_l = 0\text{ V}$), the dimensionless linear and nonlinear natural frequencies (defined in Eq. 84) of the plate are investigated.

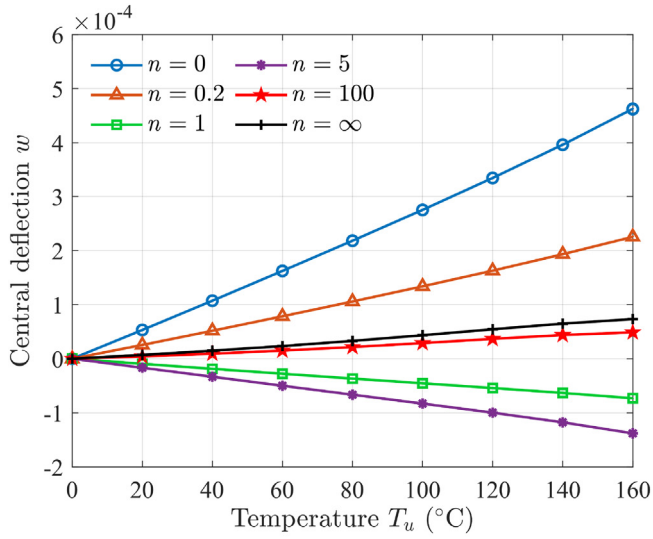
$$\bar{\omega}_{L/NL} = \frac{\omega_{L/NL} a^2 \sqrt{\rho_{m0}(1 - \nu_{m0}^2)/E_{m0}}}{h_f}, \quad (84)$$

where $\omega_{L/NL}$ denotes the linear or nonlinear natural frequency (rad/s) with considering the effect of temperature; ρ_{m0} , ν_{m0} , E_{m0} are the mass density, Poisson's ratio and Young's modulus of the SUS304 at the room temperature ($T_0 = 300\text{ K}$). Table 4 reports the calculated results of the dimensionless frequencies of the plate for different values of the temperature. It is obviously that the present IGA results for dimensionless natural frequencies match well with those derived from using FEM based on HSDT given in Ref. [26], reflecting, once again, the accuracy of the proposed formulation.

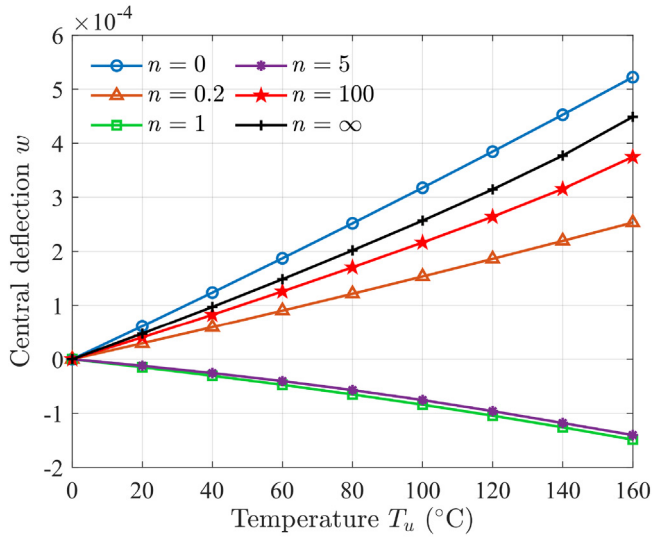
5.1.3. Example #3: Nonlinear static bending analysis

To manifest the accuracy of this study for nonlinear static bending analysis, we next consider a SSSS square FGP made of ZrO_2 and Al. The length and thickness of the plate are designed as 200 mm and 10 mm, respectively. The material properties are: for ZrO_2 , $E = 151.0\text{ GPa}$, $\nu = 0.3$, $\rho = 3000\text{ kg/m}^3$, $k_f = 2.09\text{ W/mK}$, $\alpha_f = 10 \times 10^{-6}/\text{K}$; for Al, $E = 70.0\text{ GPa}$, $\nu = 0.3$, $\rho = 2707\text{ kg/m}^3$, $k_f = 204\text{ W/mK}$, $\alpha_f = 23 \times 10^{-6}/\text{K}$.

It is assumed that the plate is subjected to a uniformly distributed load q_0 from 0 to $-1 \times 10^7\text{ N/m}^2$ with five steps. Regardless of the effect of temperature, Fig. 2 plots the variation of the dimensionless



(a) S1 structure



(b) S2 structure

Fig. 9. Nonlinear static thermal bending of PFGPs.

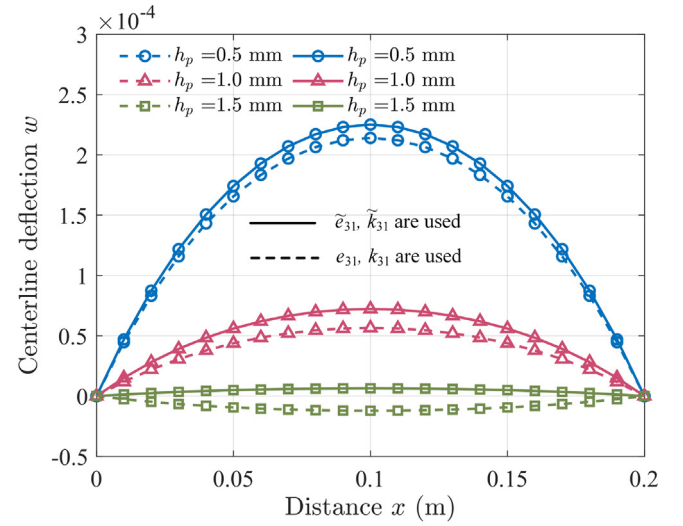
nonlinear central deflections $\bar{w} = w/h_f$ with the load parameter $\bar{P} = q_0 a^4 / E_m h_f^4$. Also, Fig. 3 depicts the computed results when the upper and lower surfaces temperatures are $T_u = 300^\circ\text{C}$ and $T_m = 20^\circ\text{C}$. Both are based on IGA, one can see that although one degree of freedom is saved in the S-FSDT, the dimensionless central deflection results are in good agreement with the existing reference solutions derived from the HSDT [50].

5.1.4. Example #4: Nonlinear transient dynamic analysis

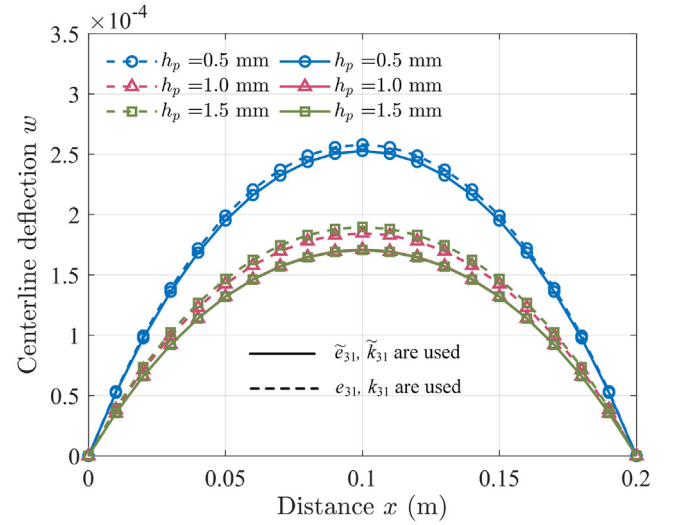
The nonlinear dynamic response of the CCCC PFGP, as the same Example#2, under the action of thermo-electro-mechanical loads is investigated. It is assumed that $n = 2$ and the mechanical load is specified as

$$f_s = q_0 \sin\left(\frac{2\pi t}{0.1}\right), \quad (85)$$

where $q_0 = 2 \times 10^9 \text{ N/m}^2$.



(a) S1 structure



(b) S2 structure

Fig. 10. The effect of the temperature dependence of the piezoelectric constants and the thickness of piezoelectric layers on the nonlinear thermal bending response.

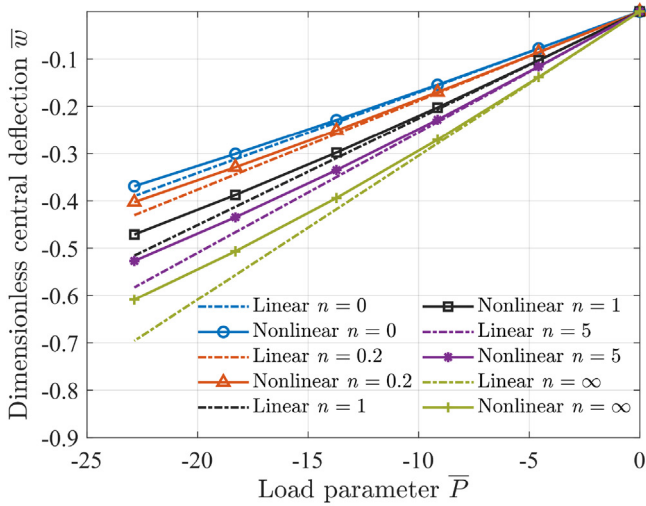
The central deflection of the plate under electrical load $V_u = V_l = 200 \text{ V}$ and temperature rise $T_u = 600 \text{ K}$, $T_l = 300 \text{ K}$ is depicted in Fig. 4. As expected, the results obtained by the present S-FSDT based IGA are in good agreement with the reference solutions employing the HSDT based FEM reported in Ref. [26].

5.2. Parametric studies

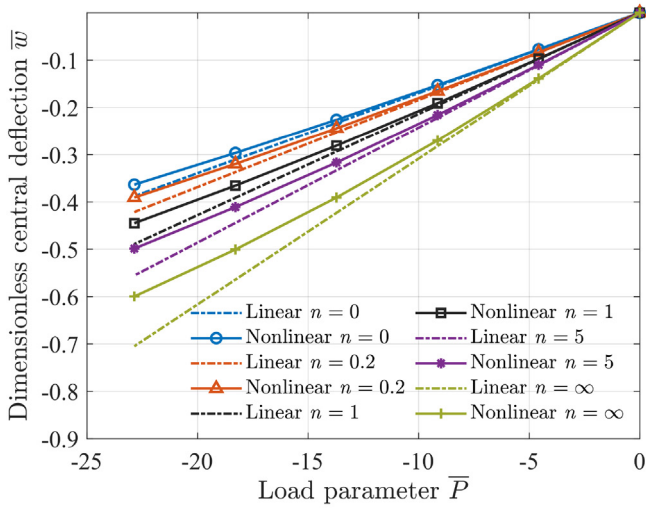
5.2.1. Thermal effects of piezoelectric stress and dielectric constants of PZT-5H

To analyze the mechanical behavior of PFGPs closer to the actual situation in thermal environment, the augmented piezoelectric constitutive Eq. (26) is now used to investigate the thermo-electro-mechanical nonlinear response of S1 and S2 structures.

From Eq. (25), the temperature-dependent piezoelectric stress constant $\tilde{e}_{31}$ and dielectric constant $\tilde{k}_{33}$ also can be defined as



(a) S1 structure



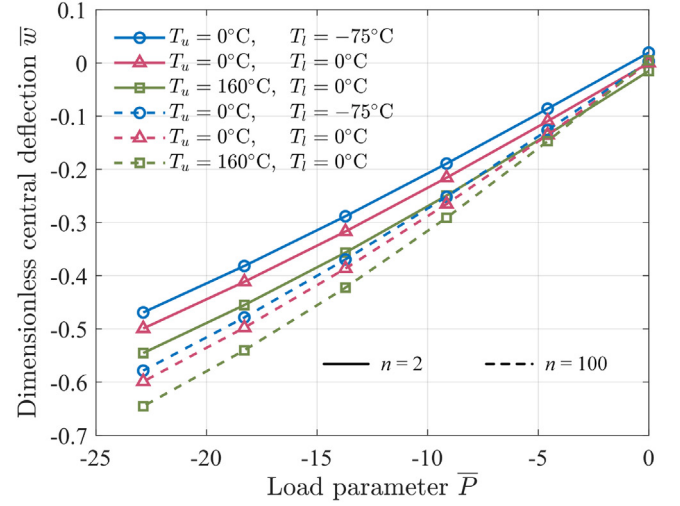
(b) S2 structure

Fig. 11. Dimensionless linear and nonlinear central deflections of the plate only subjected to the mechanical uniformly distributed load ($T_u = T_l = 0^\circ\text{C}$, $V_u = V_l = 0\text{ V}$).

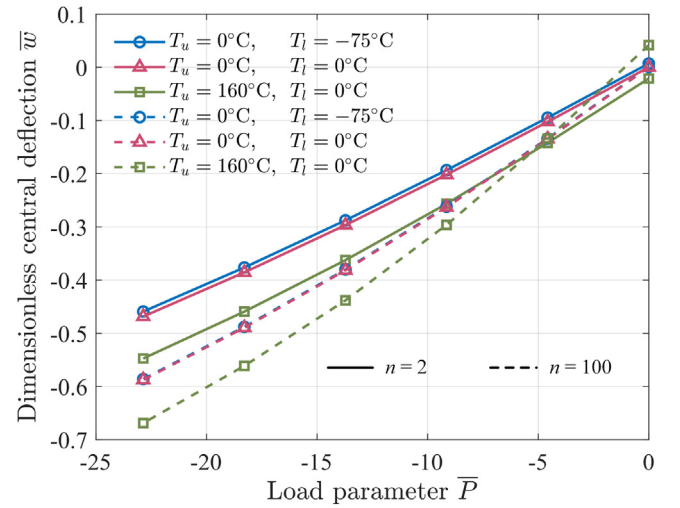
$$\tilde{e}_{31} = e_{31} + \dot{e}_{31}\Delta T, \quad \tilde{k}_{33} = k_{33} + \dot{k}_{33}\Delta T, \quad (86)$$

where e_{31} and k_{33} are the piezoelectric stress and dielectric constants at a reference temperature $T_0 = 0^\circ\text{C}$.

Here, PZT-5H is considered as the piezoelectric material because the temperature-dependent piezoelectric strain constant $\tilde{d}_{31}$ and dielectric constant $\tilde{k}_{33}/\epsilon_0$ have been measured by Wang et al. [35], where $\epsilon_0 = 8.85 \times 10^{-12}$ farad/m. Using the curve fit method, Fig. 5 shows the trend of $\tilde{d}_{31}$ and $\tilde{k}_{33}/\epsilon_0$ with temperature. The piezoelectric stress constant $\tilde{e}_{31}$ can then be calculated through Eq. (24). Figs. 6(a) and 6(b) show the variations of piezoelectric stress constant $\tilde{e}_{31}$ and dielectric constant $\tilde{k}_{33}$ with temperature. It can be seen that $\tilde{d}_{31}$ and $\tilde{e}_{31}$ varies linearly with temperature, whereas $\tilde{k}_{33}$ varies non-linearly with temperature. Using the expression forms of Eq. (86), they are expressed as



(a) S1 structure



(b) S2 structure

Fig. 12. Dimensionless nonlinear central deflections of the plates subjected to the mechanical and different thermal loads ($V_u = V_l = 0\text{ V}$).

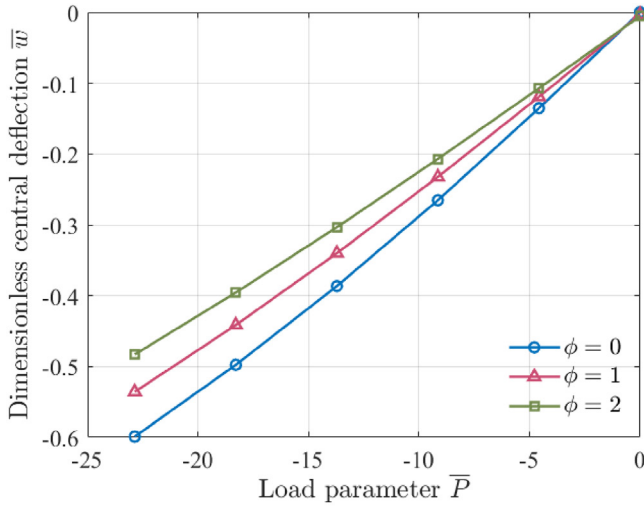
$$\tilde{e}_{31} = -34.1599 - 0.1053\Delta T, \quad (87)$$

$$\begin{aligned} \tilde{k}_{33} = & 2.717835 \times 10^{-8} + 1.3629 \times 10^{-10}\Delta T + 2.43375 \times 10^{-13}\Delta T^2 \\ & - 7.08 \times 10^{-15}\Delta T^3 - 4.2834 \times 10^{-17}\Delta T^4 + 7.2039 \times 10^{-19}\Delta T^5 \end{aligned} \quad (88)$$

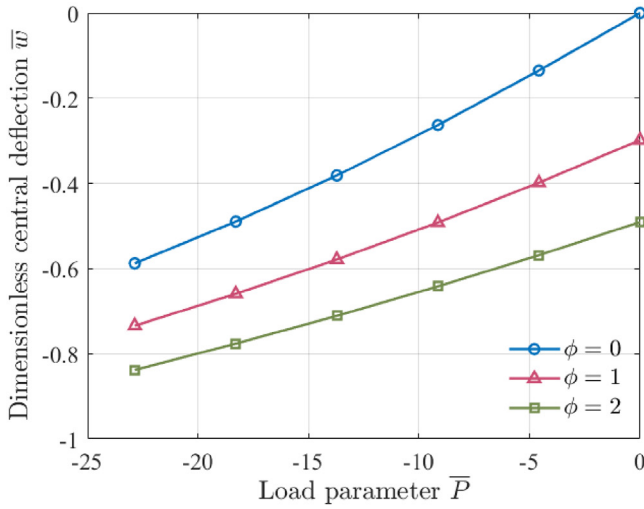
In which, -34.1599 and 2.717835×10^{-8} are the piezoelectric stress and dielectric constants at a reference temperature $T_0 = 0^\circ\text{C}$. It is important to point out here that the above relationship is only valid for PZT-5H. For other piezoelectric materials, it is necessary to re-establish a similar relationship based on the experimental data.

5.2.2. Analysis of temperature distribution for PFGPs

The effect of temperature distribution along the thickness direction is now studied. We thus consider a ZrO_2/Al FGP with surface-bonded PZT-5H piezoelectric layers. The dimensions of the plate are 200 mm for both the length and width. The thicknesses of the FGP and each piezoelectric layer are 10 mm and 0.5 mm, respectively. Other mate-



(a) S1 structure

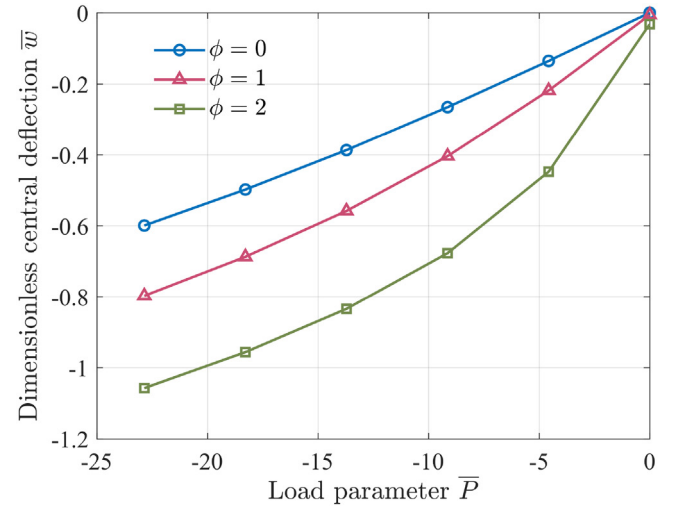


(b) S2 structure

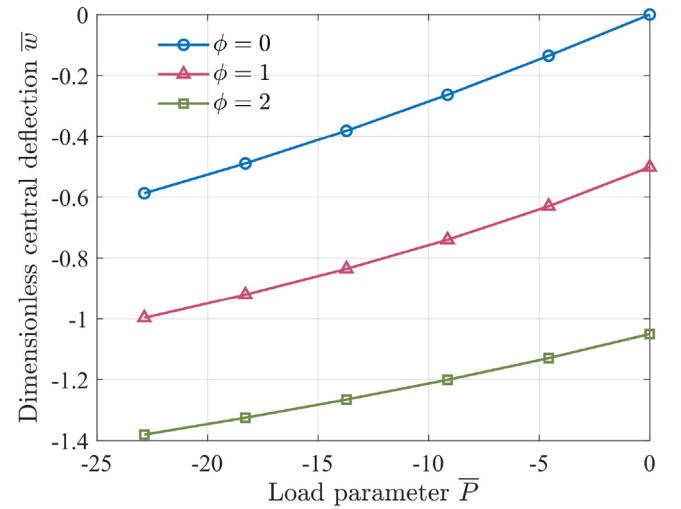
Fig. 13. Dimensionless nonlinear central deflections of the plates subjected to the electro-mechanical loads.

rial properties of PZT-5H are assumed to be temperature-independent as $E = 63.0$ GPa, $\nu = 0.30$, $\rho = 7500$ kg/m³, $\alpha_p = 5 \times 10^{-9}$ /°C, $k_p = 2.1$ W/mK, $P_3 = 4 \times 10^{-4}$ C/m² °C.

We assume that the top surface of the plate is exposed to a temperature of 160 °C and the lower surface is by a temperature of 0 °C. The results calculated for the temperature distribution through the thickness of S1 and S2 structures are visualized in Fig. 7. The comparison with the FEM solution using ANSYS software (in Ansys models, we divide each piezoelectric layer and FG layers into two and forty finite sublayers along the thickness direction and the 3D 20-node solid 90 element with thickness of 0.25 mm is adopted) proves the accuracy of the temperature distribution analysis of PFGPs. It can be observed that the upper piezoelectric layer in S1 structure is exposed to high temperature, whereas the two piezoelectric layers in S2 structure are in a relatively low temperature environment. Due to the large difference of thermal conductivity between ZrO₂ and Al, the temperature



(a) S1 structure



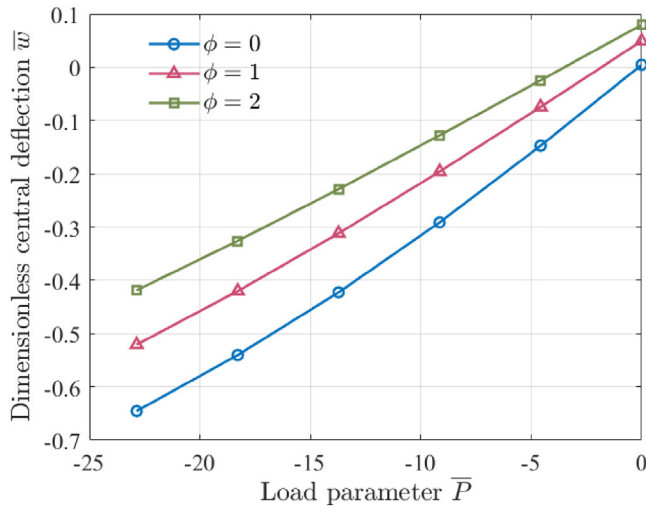
(b) S2 structure

Fig. 14. Dimensionless nonlinear central deflections of the plates subjected to the electro-mechanical loads without considering the initial stiffness matrix K_0^p .

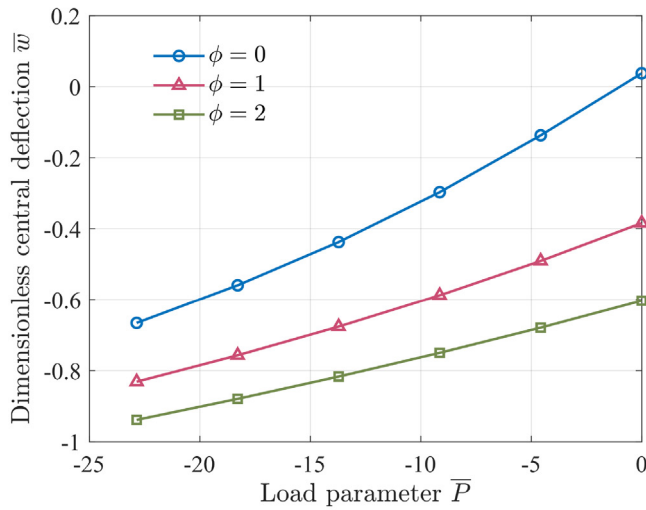
distribution is greatly dependent on the gradient index n . Because the thermal conductivity k_f of FGP increases with increasing the index n , the temperature of the lower piezoelectric layer also increases. Compared to S1 structure, the FGP in S2 structure is always in a higher temperature area.

The influence of the thickness of piezoelectric layers on the temperature distribution is also investigated. As shown in Fig. 8, the temperature of piezoelectric layers increase with its thickness. Meanwhile, one can also see that the temperature of the lower piezoelectric layers in two structures are identical, but the temperature of the upper piezoelectric layer in S2 structure is significantly smaller than that of S1 structure.

The above analysis indicates that, as compared with S1 structure, S2 structure can effectively reduce the temperature on piezoelectric materials. In terms of the selection of FGPs, if the thermal conductivity of ceramic and metal is close, the influence of gradient index n on



(a) S1 structure



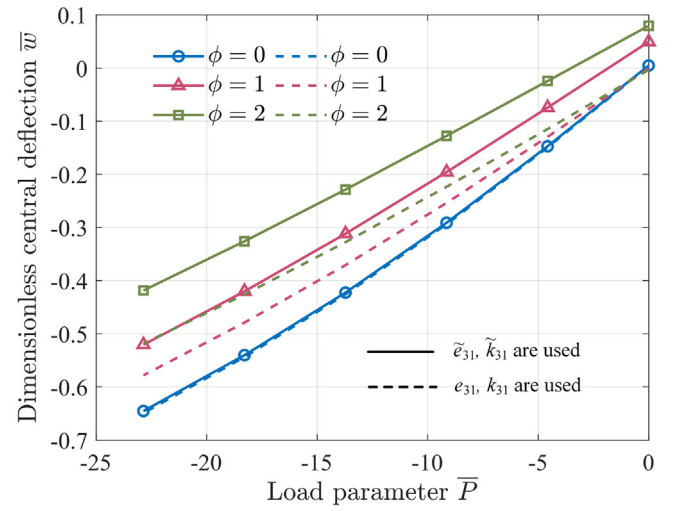
(b) S2 structure

Fig. 15. Dimensionless nonlinear central deflections of the plates subjected to thermo-electro-mechanical loads ($T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$).

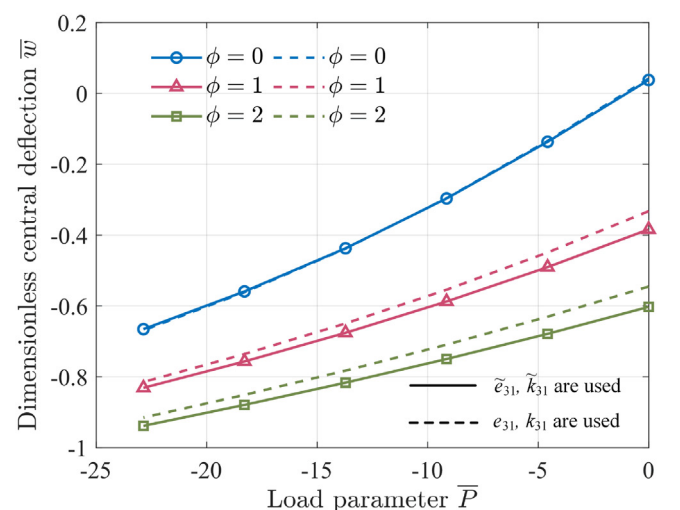
the temperature distribution through the thickness of the plates can be reduced.

5.2.3. Nonlinear free vibration

After the analysis of temperature distribution, the former two plate structures with CCCC boundary condition are used to investigate the linear and nonlinear natural frequencies of structures in thermal environment. The thermal effect of piezoelectric stress and dielectric constants on the dimensionless frequencies of the S1 and S2 structures with no electrical loads ($V_u = V_l = 0\text{ V}$) are estimated and their computed results are then given in [Tables 5 and 6](#). Without accounting for the thermal effect of piezoelectric coefficients, one can see that (a) the frequencies decrease with increasing the gradient index n ; (b) due to the nonlinear stiffness matrix, the nonlinear frequencies are higher than linear ones; (c) when n is small, the frequencies of S2 structures is greater than that of S1 structure at $T_u = T_l = 0^\circ\text{C}$, but this phenomenon reverses when n increases to a certain value; (d) because



(a) S1 structure



(b) S2 structure

Fig. 16. Dimensionless deflections of the plates with and without considering the temperature influence of the piezoelectric stress and dielectric constants ($T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$).

of the existence of the initial thermal stress stiffness matrix K_0^{th} , the frequencies decrease with increasing the temperature; and (e) the effect of temperature gradient on the frequencies of S2 structure is greater than that of S1 structure due to the temperature of FGP in S2 structure, which is much higher than that in S1 structure. In addition, the frequencies with considering the thermal effect of piezoelectric stress and dielectric constants in thermal environment are greater than that without accounting for the temperature effect. Therefore, the augmented piezoelectric constitutive equation used in this investigation is conducive to analyze the mechanical behavior of PFGPs more closer to the facts in thermal environment.

The frequencies of the S1 and S2 structures with different electric loads at temperature of $T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$ are presented in [Tables 7 and 8](#). The numerical results indicate the frequencies of the plates, which increase when positive voltage is applied, whereas decreases when negative voltage is imposed.

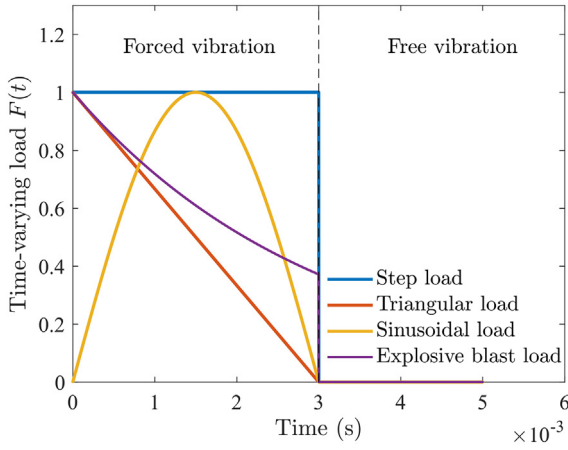


Fig. 17. Four different distributed transverse forces.

5.2.4. Nonlinear static bending analysis

The geometrically nonlinear static bending response of two PFGP structures subjected to thermo-electro-mechanical loads is studied. The boundary conditions of the plates are set to be SSSS.

(a). Nonlinear thermal bending response of PFGPs

It is assumed that the temperature of the lower surface of the plates is set to 0 °C. The variation of the nonlinear central deflections with the thermal load is plotted in Fig. 9. The temperature greatly alters the deflections as they increase with increasing the temperature. The thermal nonlinear deflections of the S2 structure is larger than that of the S1 structure due to the different positions of the piezoelectric layers in the two structures. When $n = 0$, the FPGs are composed of ceramic, and the temperature of the upper surface of the plates is higher than that of the lower surface. Hence, the center of the plate is inflected upward. With increasing the gradient index n , the metal components in FPGs increase, and the temperature distribution also shows a great change as sketched in Fig. 7. The thermal expansion coefficient of metal is much larger than that of ceramic, resulting in the decrease of the central deflections of the plates from upward deflections to downward deflections. When n increases to a certain extent (e.g., $n = 100$, $n = \infty$), the FPGs are almost or completely composed of metal, and the deflections appear positive again.

To study the effect of the temperature dependence of the piezoelectric constants and the thickness of piezoelectric layers on the nonlinear thermal bending response of the plates, Fig. 10 shows the centerline deflection of the plates with $n = 0.2$ at temperature $T_u = 160$ °C, $T_l = 0$ °C. One can see that for S1 structure, the deflection considering the thermal effect of piezoelectric constants is larger than that without

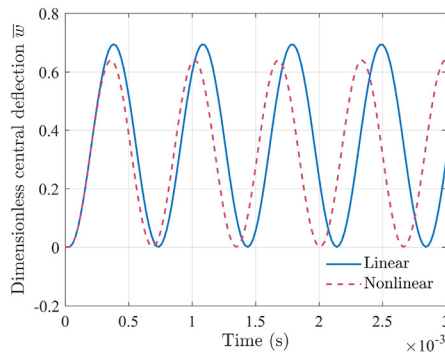
considering. However, for S2 structure, the results are opposite. Besides, this difference increases with the increase of the thickness of the piezoelectric layers.

(b). Nonlinear static bending response of PFGPs under thermo-electro-mechanical loads

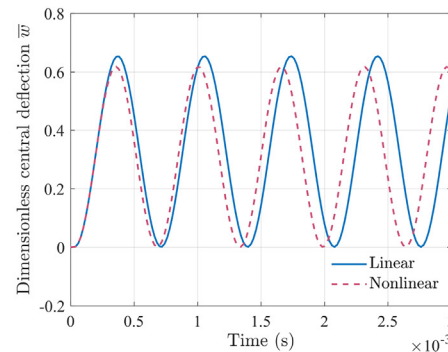
Similar to Example #3, a uniformly distributed load of -1×10^7 N/m² with five steps is used for investigating the nonlinear static bending response of the plates under mechanical, thermo-mechanical, electro-mechanical, and thermo-electro-mechanical loads. Fig. 11 plots the dimensionless linear and nonlinear central deflections of the plate only subjected to the uniformly distributed load. As expected, the deflections increase with increasing the gradient index n . The nonlinear deflections are smaller than linear deflections due to the existence of nonlinear stiffness matrix. Additionally, it can be observed that except when $n = \infty$, the obtained deflections of the S1 structure are greater than those of the S2 structure, which is also consistent with the results of nonlinear free vibration analyzed in SubSection 5.2.3.

The dimensionless nonlinear central deflections of the plates subjected to the mechanical and different thermal loads with $n = 2$ and $n = 100$ are also calculated and given in Fig. 12. The deflections are not zero when the mechanical load is zero because there is a temperature difference between the upper and lower surfaces of the plates. The magnitude and direction of the deflections completely caused by the thermal loads (when the mechanical load is equal to zero) are related to the materials and temperature distributions of PFGPs. When the plates are subjected to both mechanical and thermal loads, one can see that the deflections decrease with decreasing the temperature and increase with increasing the temperature. Similar to the section of nonlinear free vibration analysis, the reason for this phenomenon is that the initial thermal stress stiffness matrix K_0^{th} decreases and geometric stiffness matrix K_G increases with the decrease of temperature. Conversely, the increase of temperature leads to the increase of K_0^{th} and decrease of K_G .

Next, PFGPs with $n = 100$ are used for studying the nonlinear static bending response under electro-mechanical and thermo-electro-mechanical loads. A parameter input voltage ($V_u = V_l = \phi/k_{33} \times 10^3$) is introduced because the static deflection induced by electric load is not obvious. As shown in Fig. 13, the effect of voltage on S2 structure is greater than that of S1 structure. The deflections are reduced with increasing the voltage for the S1 structure, whilst amplified as the voltage increases for the S2 structure. This is due to the fact that the deflections increment caused by the electric loads is much smaller than the reduction caused by the initial electric stiffness matrix K_0^p for S1 structure. To clarify this conclusion, Fig. 14 visualizes the deflections of the plates under electro-mechanical load without accounting for the initial electric stiffness matrix. One can see that, for two structures, the deflections increase with increasing the voltage.



(a) S1 structure



(b) S2 structure

Fig. 18. Dimensionless linear and nonlinear dynamic responses of plates only subjected to the step load ($T_u = T_l = 0$ °C, $V_u = V_l = 0$ V).

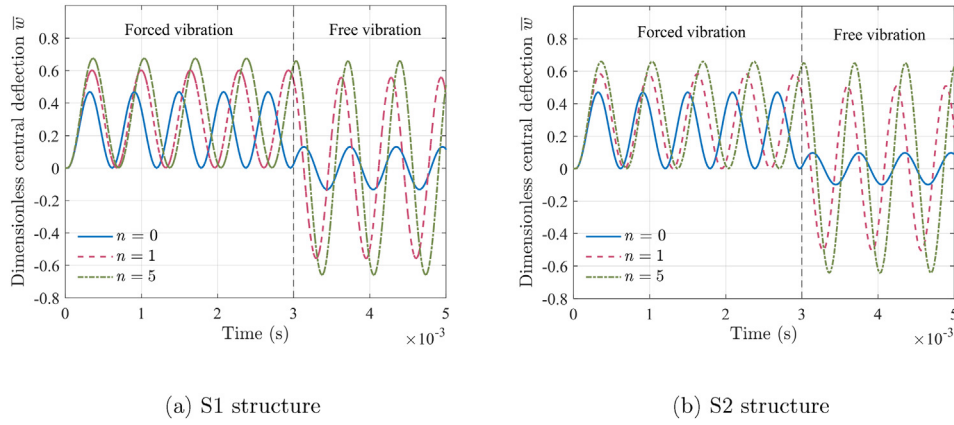


Fig. 19. The effect of the gradient index on the nonlinear dynamic response ($T_u = T_l = 0^\circ\text{C}$, $V_u = V_l = 0\text{ V}$).

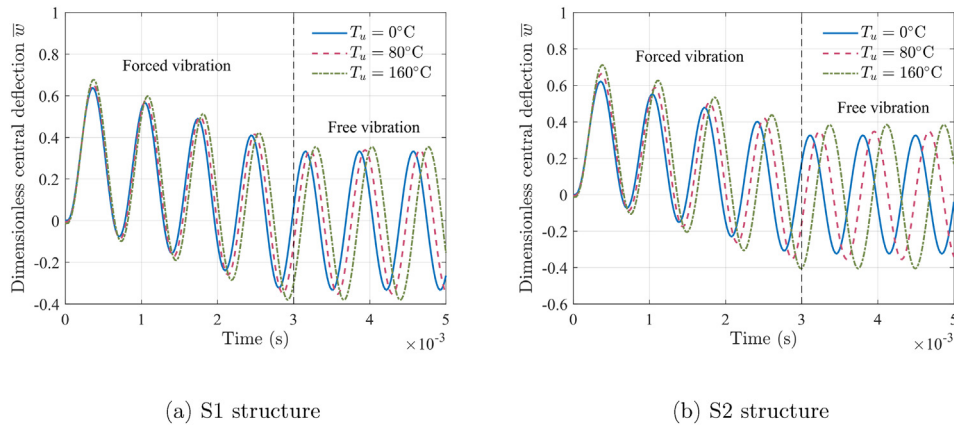


Fig. 20. Dynamic deflections of the plates subjected to triangular load and different thermal loads ($V_u = V_l = 0\text{ V}$).

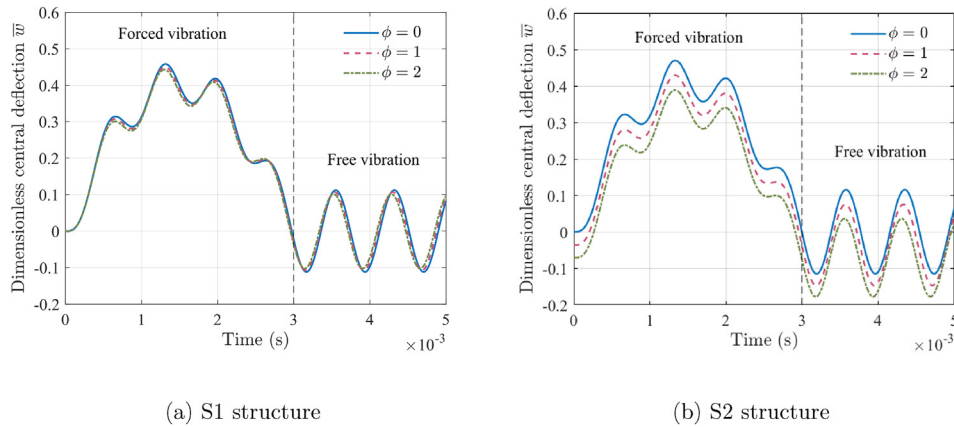


Fig. 21. The dynamic deflections of the plates with $n = 100$ subjected to sinusoidal load and different voltages at temperature $T_u = T_l = 0^\circ\text{C}$.

Therefore, the initial stiffness matrix generated by electric in-plane forces is indispensable in analysis of smart piezoelectric structures with the electro-mechanical loads.

Then, the upper and lower surfaces of the plates are assumed to be exposed to temperatures of 160°C and 0°C , respectively. Fig. 15 represents the deflections of the plates with different parameter input voltages. Compared with Fig. (13), it can be observed that the deflections of S2 structure increase because the introduction of thermal loads. However, the deflections of S1 structure are reduced, when

the mechanical, electrical and thermal loads are applied simultaneously. To investigate the major cause of this phenomenon, the deflections of the plates with and without considering the temperature influence of the piezoelectric stress constant ($\bar{e}_{31}, e_{31}$) and dielectric constant ($\bar{k}_{33}, k_{33}$) are studied. As shown in Fig. 16, the deflections of S1 structure with considering the temperature influence of the piezoelectric stress constant $\bar{e}_{31}$ and dielectric constant $\bar{k}_{33}$ are smaller than that without considering the temperature effect (e_{31}, k_{33}). Hence, we infer that the deflections of $\phi = 1$ and $\phi = 2$ in Fig. 14(a) are smaller

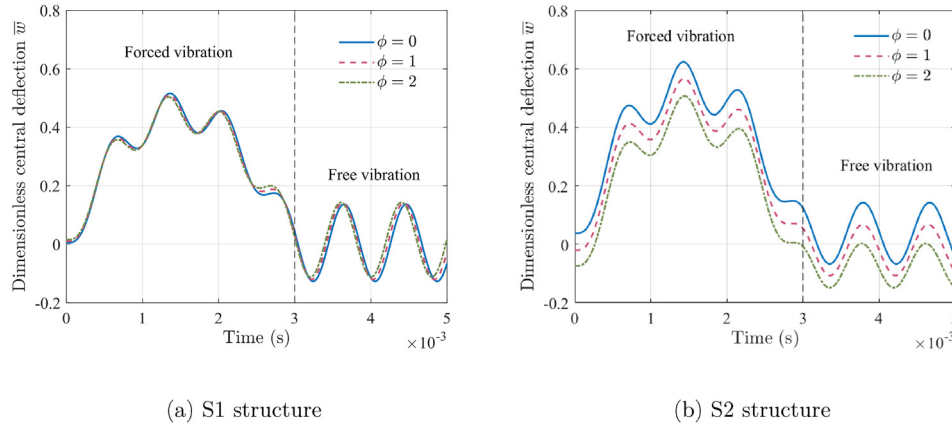


Fig. 22. The dynamic deflections of the plates with $n = 100$ subjected to sinusoidal load and different voltages at temperature $T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$.

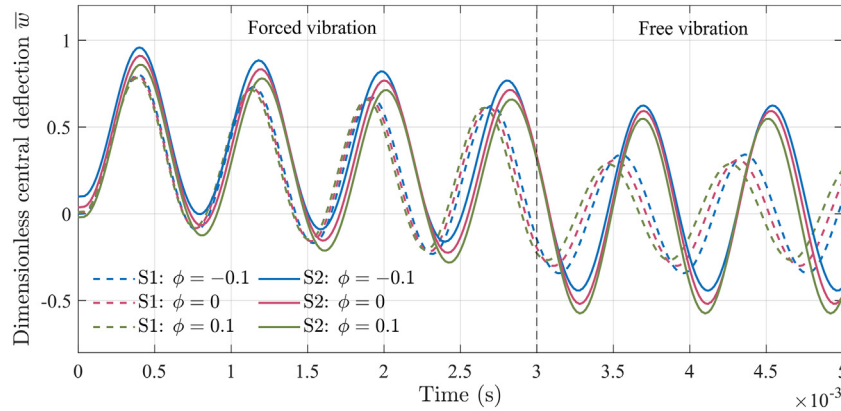


Fig. 23. The dynamic deflections of the plates with $n = 100$ subjected to explosive load and different voltages at temperature $T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$.

than those in Fig. 12(a) because of the effect of temperature on piezoelectric parameters. However, due to the difference in structure, the deflections of S2 structure with considering the temperature influence of the piezoelectric constants are larger than that without considering the temperature effect. In addition, the difference between considering and not considering the temperature effect of S2 structure is smaller than that of S1 structure. This proves that S2 structure is more conducive to maintaining the stability of the smart structures.

5.2.5. Nonlinear transient dynamic analysis

The last study deals with nonlinear transient dynamic behavior. As depicted in Fig. 17, four different distributed transverse forces defined in Eq.(89) are used for studying the nonlinear dynamic responses of PFGPs.

$$q = q_0 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{b}\right) F(t), \quad (89)$$

where $q_0 = 10\text{MPa}$ and $F(t)$ is defined as

$$F(t) = \begin{cases} \begin{cases} 1 & \text{Stepload} \\ 1 - t/t_1 & \text{Triangular load} \\ \sin(\pi t/t_1) & \text{Sinusoidal load} \\ e^{-\bar{\gamma} t} & \text{Explosive load} \end{cases} & \text{if } 0 \leq t \leq t_1 \\ 0 & \text{if } t > t_1 \end{cases} \quad (90)$$

in which $\bar{\gamma} = 330\text{s}^{-1}$ and $t_1 = 0.003\text{s}$.

The computed dimensionless linear and nonlinear dynamic responses of plates with $n = 2$ only subjected to the step load are then plotted in Fig. 18. It can be observed that the nonlinear deflections are less than the linear deflections, but the nonlinear vibration has higher frequencies. The vibration amplitudes of S1 structure are larger than those by the S2 structure. The effect of the gradient index n on the nonlinear dynamic response is represented in Fig. 19. The computed results indicate the significant variation of the response as that the stiffness of the plates decrease with increasing the index n .

The temperature of the lower surface of the plates are maintained to be 0°C . The dynamic deflections of the plates with $n = 5$ subjected to triangular load and different thermal loads are investigated in Fig. 20. The results show that the temperature rise increase the vibration amplitudes of the plate because the stiffness of the plates are reduced. At initial time, $t = 0$, there is an insignificant static offset induced by the temperature gradient.

The dynamic deflections of the plates with $n = 100$ subjected to sinusoidal load and different voltages at temperature $T_u = T_l = 0^\circ\text{C}$ and $T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$ are given in Figs. 21 and 22. The computed deflections decrease with increasing the voltage while increase with increasing the temperature. Obviously, The effect of voltage on S2 structure is greater than that on S1 structure. Fig. 23 plots two structures with $n = 100$ subjected to sinusoidal load and different voltages at temperature $T_u = 160^\circ\text{C}$, $T_l = 0^\circ\text{C}$. It can be seen that the negative voltage increases the amplitude of vibration, while the positive voltage decreases the amplitude.

6. Conclusions

In this contribution, we have outlined the development of the IGA integrated with the S-FSDT for assessment of geometric nonlinear free vibration, nonlinear static bending, and transient dynamic response of two different smart piezoelectric functionally graded plates (PFGPs) in thermal environment. Major conclusions drawn from the study can be summarized as follows:

- Through comparison studies, the accuracy and performance of the present S-FSDT based IGA method have been numerically confirmed. The developed approach is effective for studying the geometric nonlinear mechanical behavior of PFGPs, although there is one less variable compared to FSDT and HSDTs.
- Compared with the S1 structure, the S2 structure can effectively reduce the temperature on piezoelectric materials, which is conducive to maintaining the stability of smart structures.
- The linear and nonlinear frequencies with considering the thermal effect of piezoelectric stress and dielectric constants of PZT-5H in thermal environment are greater than that without considering the temperature effect.
- In thermal bending analysis, the difference between considering and not considering the thermal effect of piezoelectric constants increases with the increase of the thickness of the piezoelectric layers.
- The nonlinear deflections are smaller than linear deflections due to the existence of nonlinear stiffness matrix.
- The effect of voltage on S2 structure is greater than that of S1 structure.

Although the geometrically nonlinear mechanical behavior of two smart piezoelectric functionally graded plates (PFGPs) is investigated in this paper, it should be realized that the geometries are still simple. In order to ensure that the work more meaningful, highly complex smart structures should be studied. In the analysis of nonlinear forced vibration, only the time response of the structures is discussed. However, compared with the time response, the frequency response [51,52] can show the nonlinear dynamic behavior of the structures more comprehensively. Therefore, the study of frequency response for highly complex smart structures would be an interesting work for our future studies.

CRedit authorship contribution statement

Tao Liu: Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Chaodong Li:** Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Chao Wang:** Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Wenfeng Hu:** Conceptualization, Supervision. **Tinh Quoc Bui:** Conceptualization, Supervision, Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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